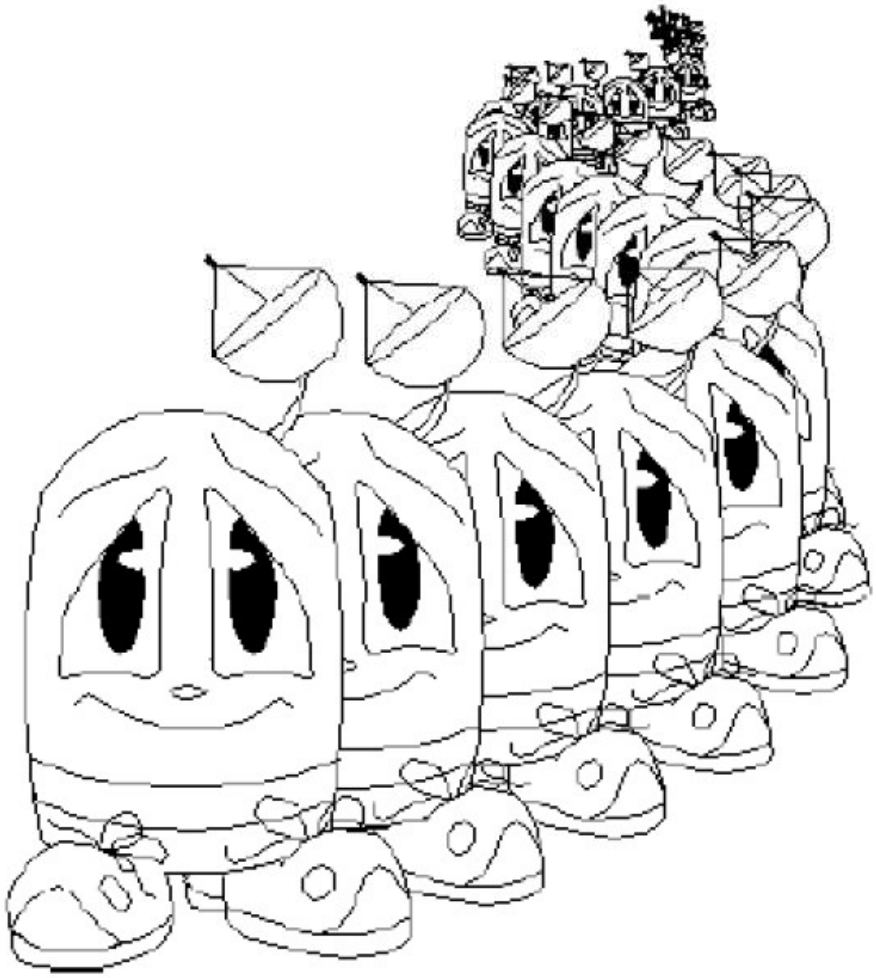
Autômato Finito Não Determinístico (AFND)



Definição: Não Determinismo

É uma importante generalização dos modelos de máquinas abstratas, na Teoria da Computação.

Numa transição não determinista, partindo-se de um estado q_1, lendo-se um símbolo a, é possível ir para um ou mais estados. Por exemplo: $\delta(q_1, a) = \{q_1, q_2\}$.

Nem sempre o não-determinismo aumenta o poder de reconhecimento de uma classe de autômatos. Por exemplo, o Autômato Finito Não Determinístico tem poder de reconhecimento equivalente ao Autômato Finito Determinístico.



Definição: Autômato Finito Não Determinístico – Def1

Um AFND é um Autômato Finito, onde são permitidas transições não determinísticas.

Uma palavra é aceita no AFND se existe alguma seqüência de transições, processando a palavra símbolo a símbolo, que leve do estado inicial a um estado final.

Na representação de máquina abstrata de reconhecimento (fita de entrada + unidade de controle finito), o AFND assume um conjunto de estados alternativos, como se houvesse uma multiplicação de unidades de controle com processamento independente



Definição: Autômato Finito Não Determinístico – Def. 2

Um Autômato Finito é uma 5-upla:

$A = (Q, \Sigma, \delta, q_0, F)$

$Q \rightarrow$ conjunto finito de estados

$\Sigma \rightarrow$ alfabeto (símbolos da fita): $\Sigma \cap Q = \varnothing$

$q_0 \rightarrow$ estado inicial

$\delta \rightarrow$ função de transição de estado: $Q \times \Sigma \rightarrow 2^Q$,

onde 2^Q é o conjunto potência de Q (todos os subconjuntos)

$F \rightarrow$ conjuntos de estados finais (aceitação): $F \subseteq Q$



Definição: Função de Transição de Estados δ (AFND)

Para todos os estados do AFND ($q \in Q$), a função deve definir quais serão os possíveis estados do autômato ($\{q_1, q_{2, \ldots}\} \subset Q$), quando for lido qualquer símbolo na fita ($a \in \sum$).

A função δ é dada por uma lista de transições do tipo:

$$\delta(q, a) = \{q_1, q_2, \ldots, q_N\},$$

ou seja,

o AFND no estado q e lendo o símbolo a, escolhe um dos estados q_i ($1 \leq i \leq N$) como novo estado.

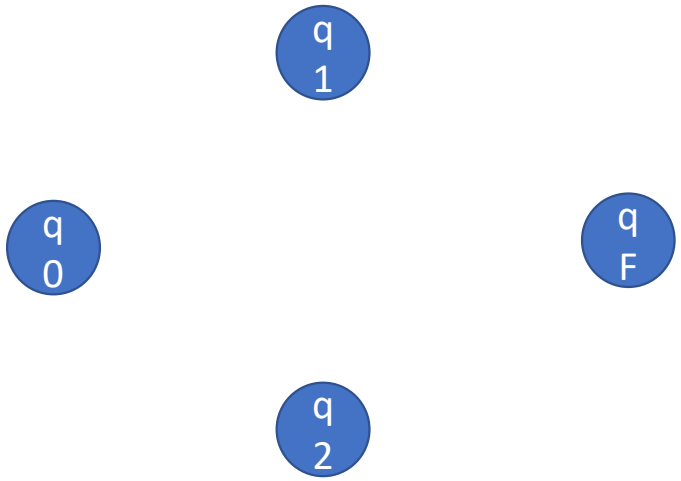


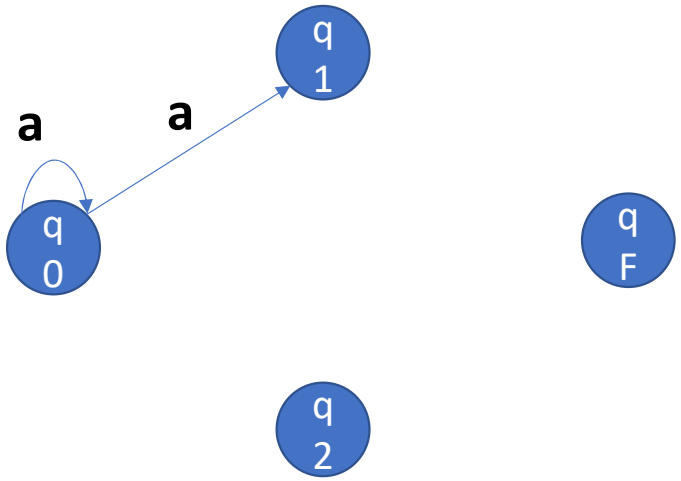
Exemplo 1:

δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

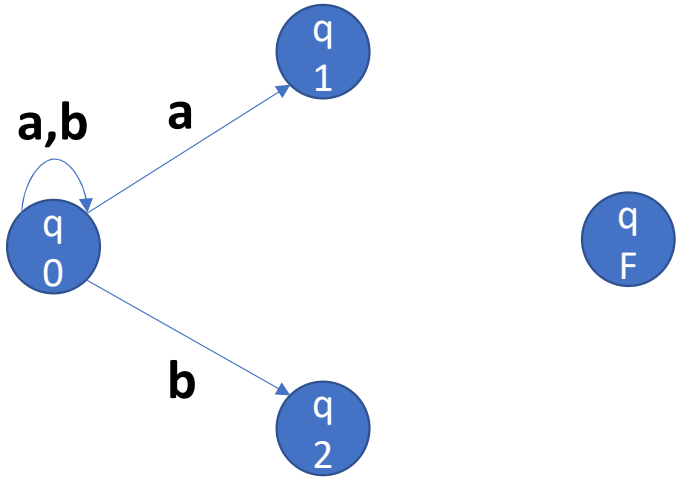
Qual á a Linguagem reconhecida/aceita por esse AF?

δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

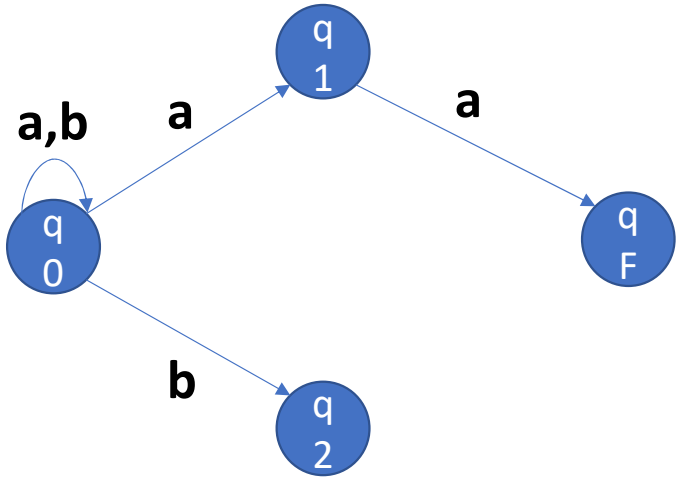




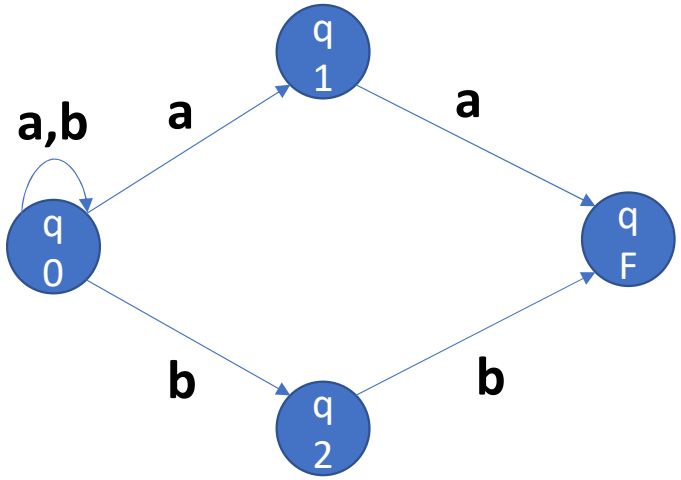
δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$



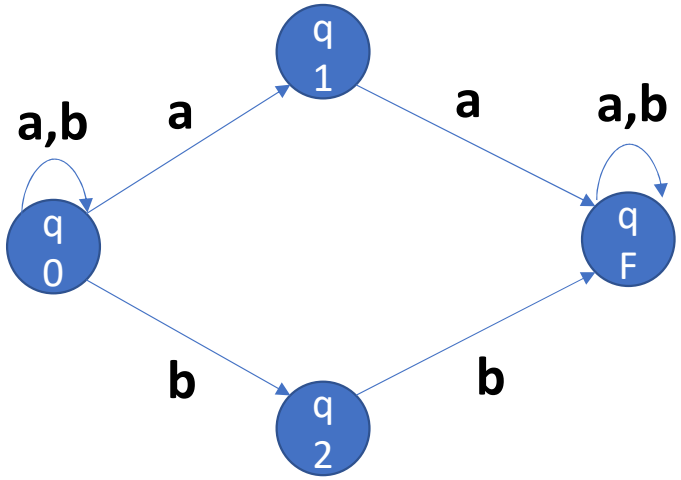
δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$



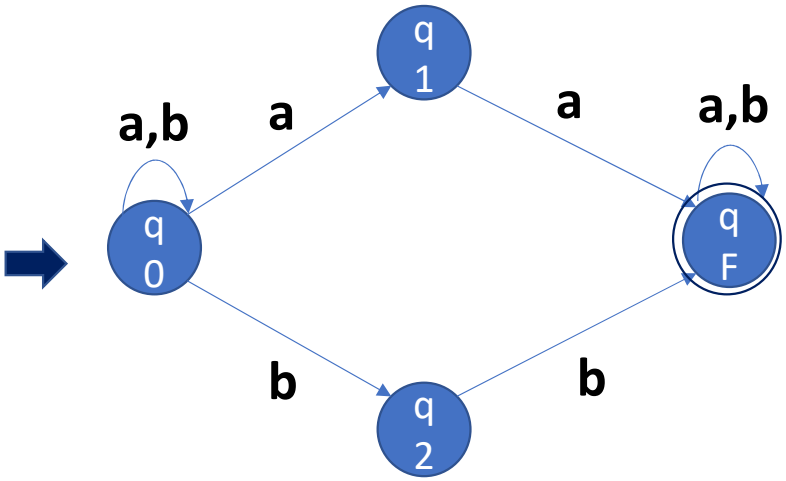
δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$



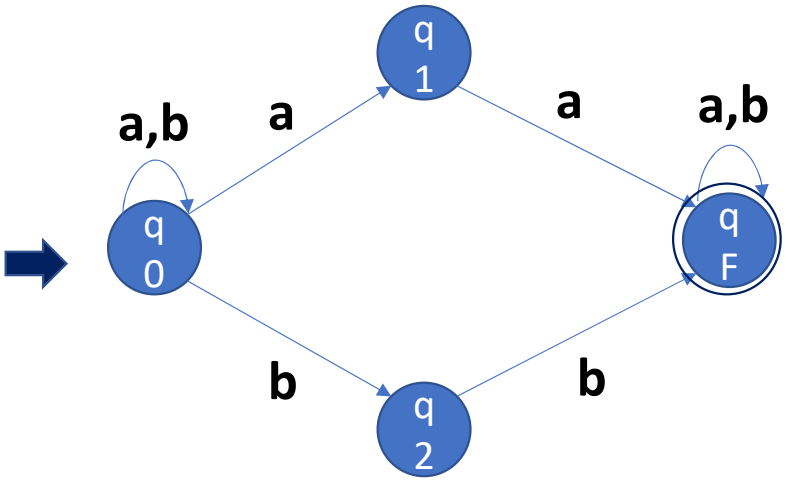
δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$



δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

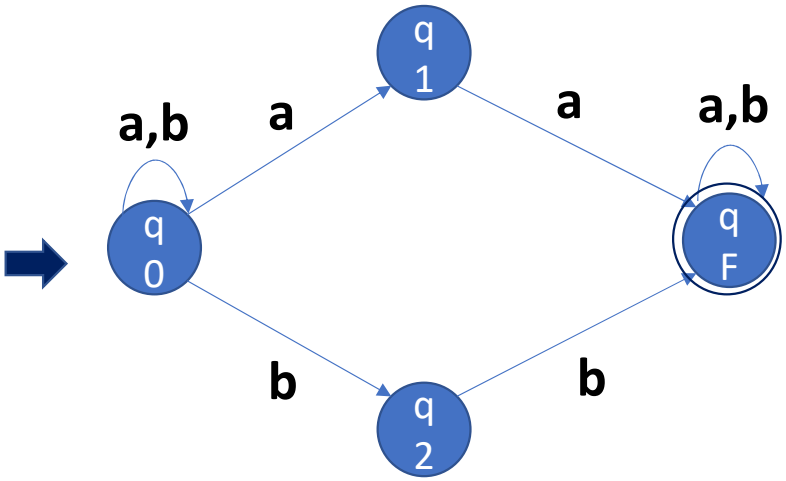


δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$



δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

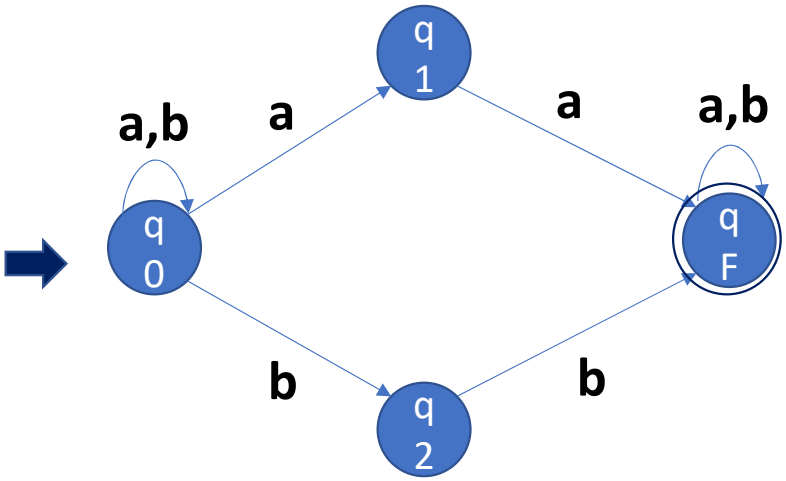
Processamento de palavras:



δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

Processamento de palavras:

aa ⟹ aceita ou rejeita?

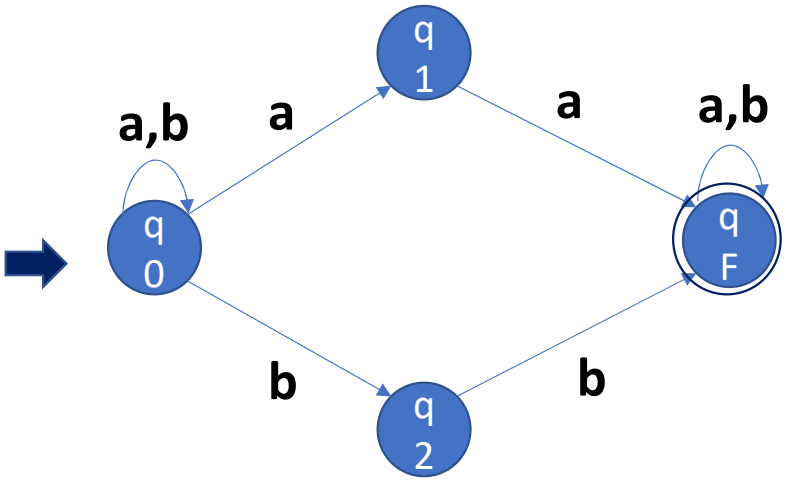


δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

Processamento de palavras:

aa ⟶ aceita

bb ⟶ ?

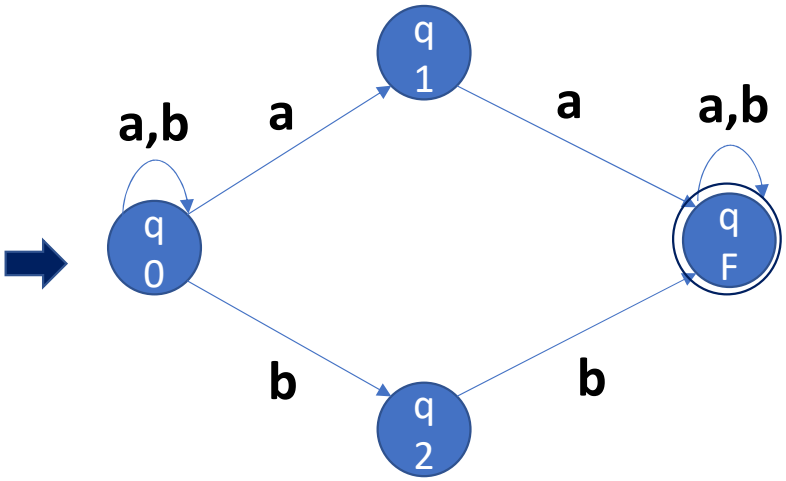


δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

Processamento de palavras:

aa, bb ⟶ aceita

ab ⟶ ?



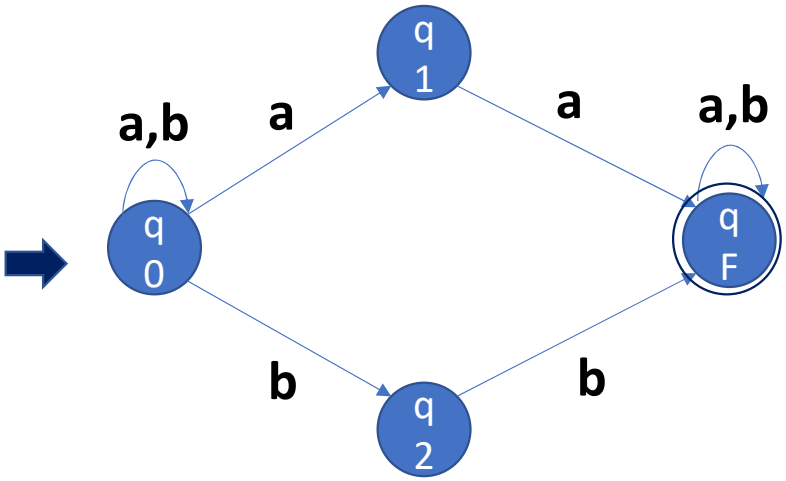
δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

Processamento de palavras:

aa, bb ⟶ aceita

ab ⟶ rejeita

baa ⟶ ?



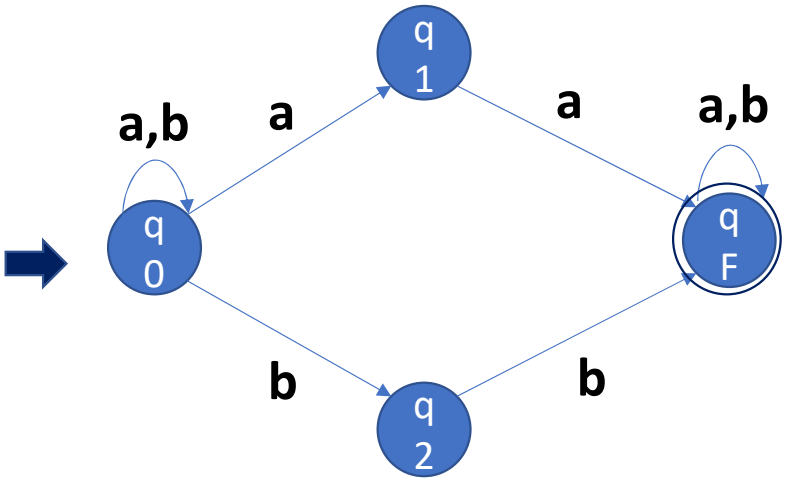
δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

Processamento de palavras:

aa, bb, baa ⟶ aceita

ab ⟶ rejeita

aba, abaa, baba, bababb ⟶ ?



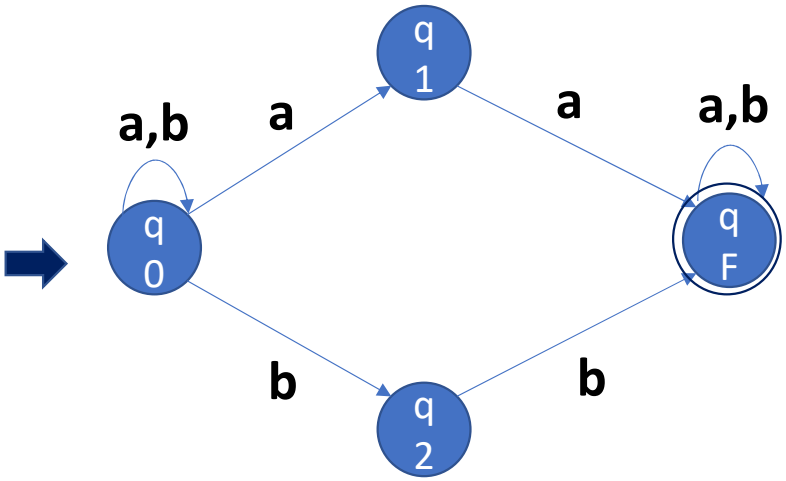
δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

Processamento de palavras:

aa, bb, baa, abaa, bababb ⟹ aceita

ab, aba, baba ⟹ rejeita

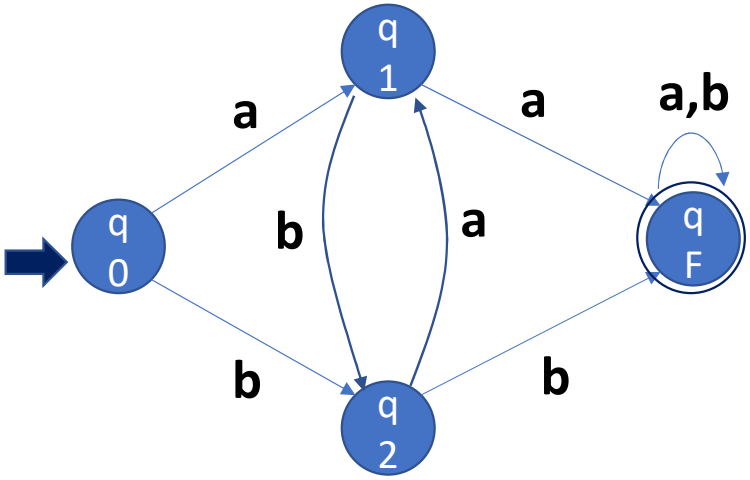
Linguagem aceita pelo AFND ?



δ_5	a	b
➡ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$

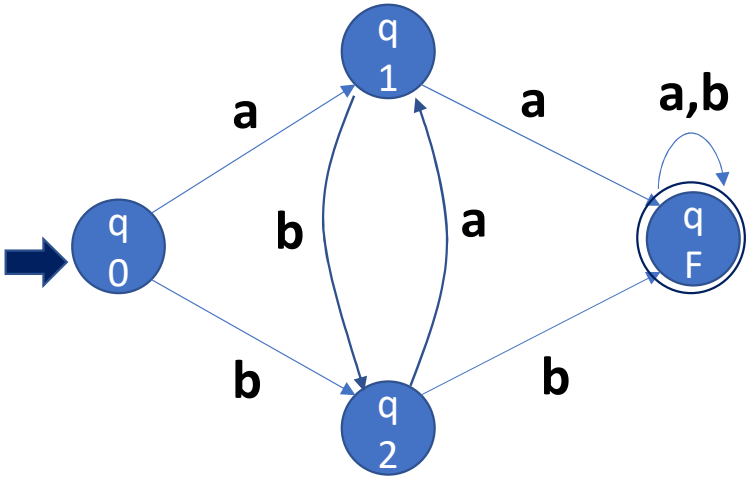
Linguagem aceita: L = { w ϵ {a,b}* | “aa” ou “bb” é subpalavra de w}

L = { w ϵ {a,b}* | “aa” ou “bb” é subpalavra de w}

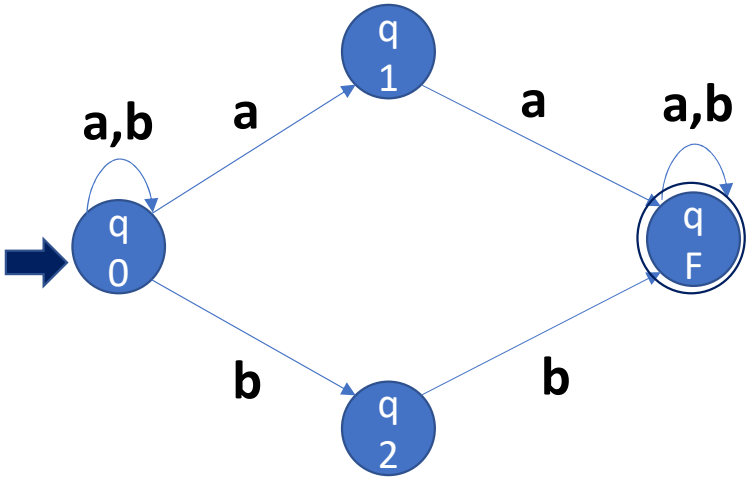


δ_1	a	b
→ q_0	q_1	q_2
q_1	q_f	q_2
q_2	q_1	q_f
★ q_f	q_f	q_f

L = { w ϵ {a,b}* | “aa” ou “bb” é subpalavra de w}



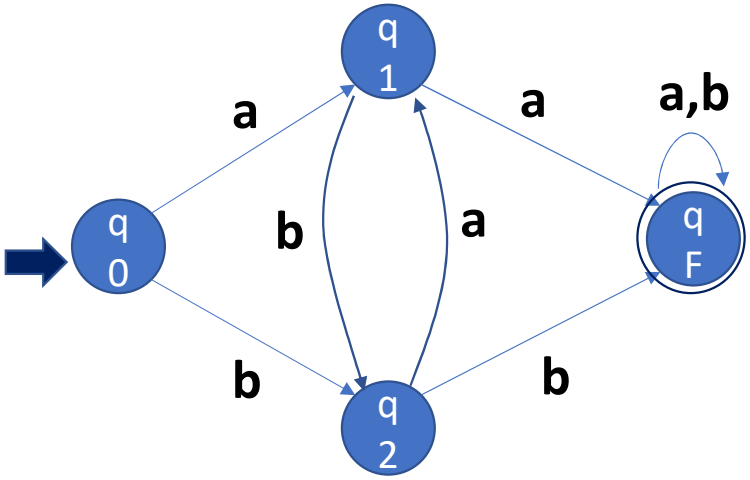
δ_1	a	b
→ q_0	q_1	q_2
q_1	q_f	q_2
q_2	q_1	q_f
★ q_f	q_f	q_f

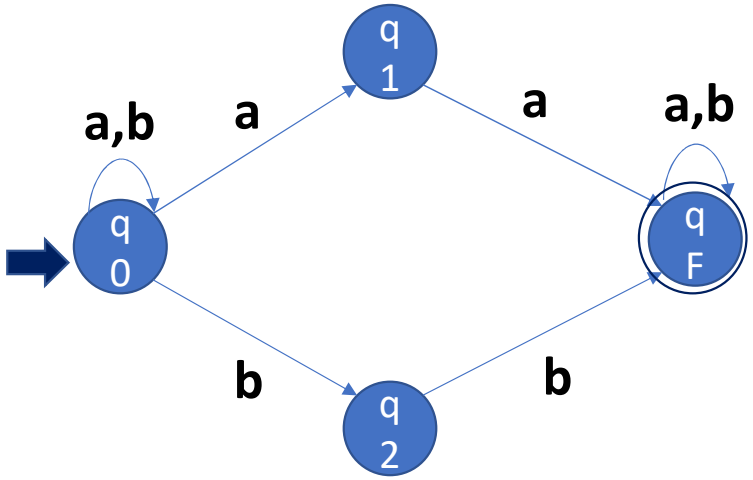


δ_5	a	b
→ q_0	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_f\}$	-
q_2	-	$\{q_f\}$
★ q_f	$\{q_f\}$	$\{q_f\}$



L = { w ϵ {a,b}* | “aa” ou “bb” é subpalavra de w}





Exercício 1 (para entregar):

Elaborar um AFD e um AFND que reconheçam as seguintes linguagens:

a) L2 = { w ϵ {a,b}* | “aa” ou “bb” é prefixo de w}
b) L3 = { w ϵ {a,b}* | “aa” ou “bb” é sufixo de w}

Exemplo 2:

- $M_6 = (\{a, b\}, \{q_0, q_1, q_2, q_f\}, \delta_6, q_0, \{q_f\})$

Linguagem reconhecida/aceita por esse AF?

L = { w ∈ {a,b}* | “aaa” é sufixo de w}



Exemplo 2:

- $M_6 = (\{a, b\}, \{q_0, q_1, q_2, q_f\}, \delta_6, q_0, \{q_f\})$



Linguagem reconhecida/aceita por esse AF

L = { w ∈ {a,b}* | “aaa” é sufixo de w}

Exemplo 2:

- $M_6 = (\{a, b\}, \{q_0, q_1, q_2, q_f\}, \delta_6, q_0, \{q_f\})$

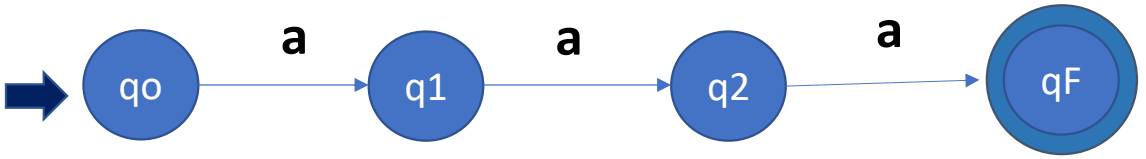


Linguagem reconhecida/aceita por esse AF

L = { w ∈ {a,b}* | “aaa” é sufixo de w}

Exemplo 2:

- $M_6 = (\{a, b\}, \{q_0, q_1, q_2, q_f\}, \delta_6, q_0, \{q_f\})$

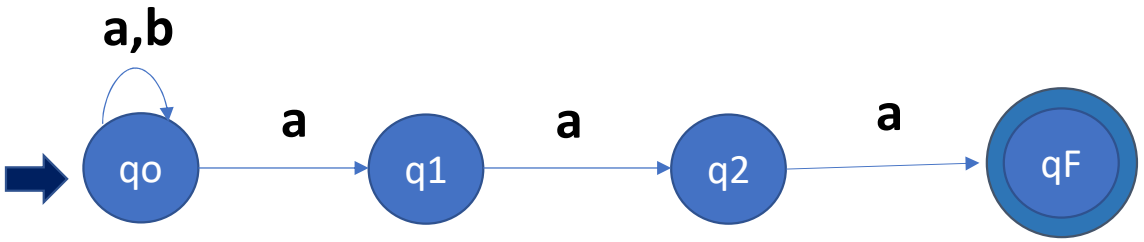


Linguagem reconhecida/aceita por esse AF

L = { w ∈ {a,b}* | “aaa” é sufixo de w}

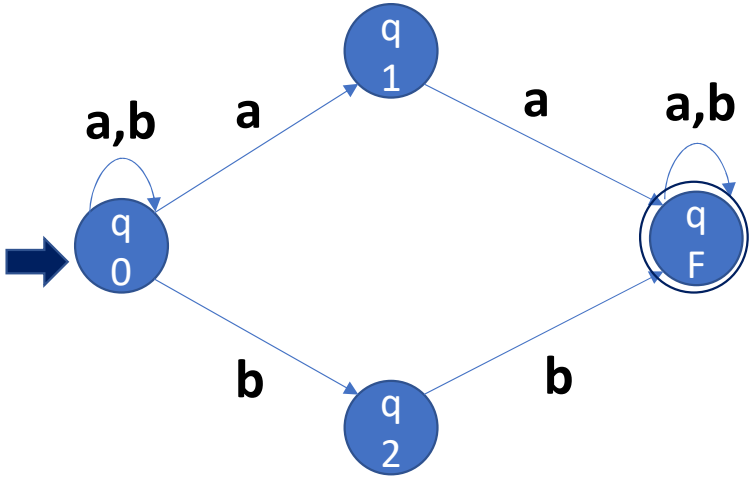
Exemplo 2:

- $M_6 = (\{a, b\}, \{q_0, q_1, q_2, q_f\}, \delta_6, q_0, \{q_f\})$



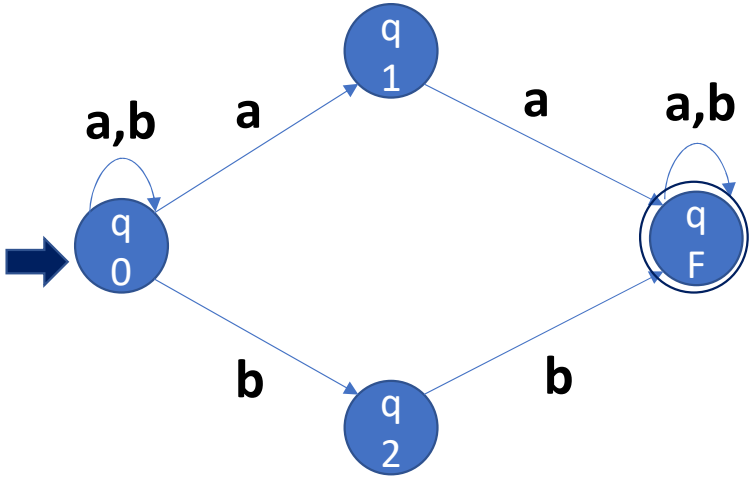
Linguagem reconhecida/aceita por esse AF

L = { w ϵ {a,b}* | “aaa” é sufixo de w}



Processamento de palavras:

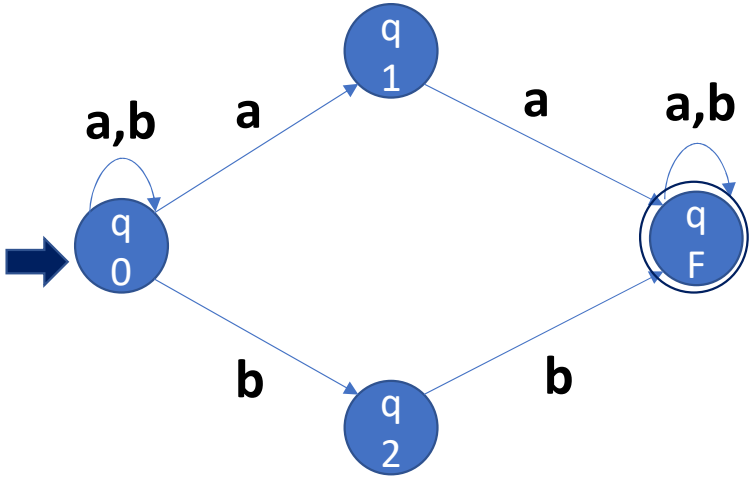
aa



Processamento de palavras:

aa

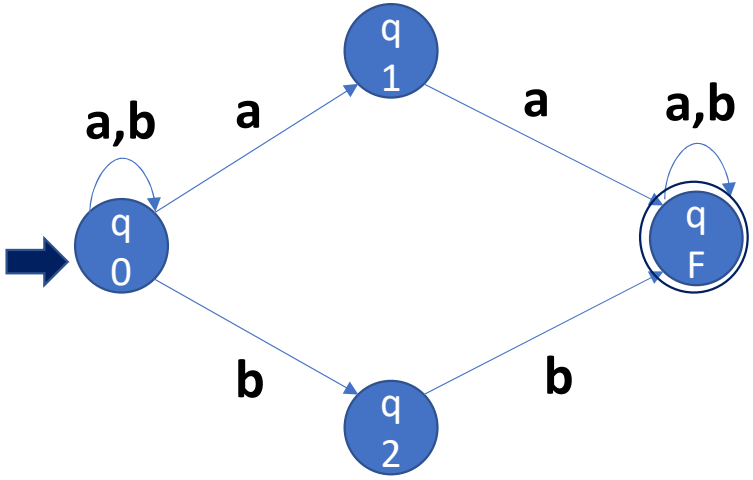
q0



Processamento de palavras:

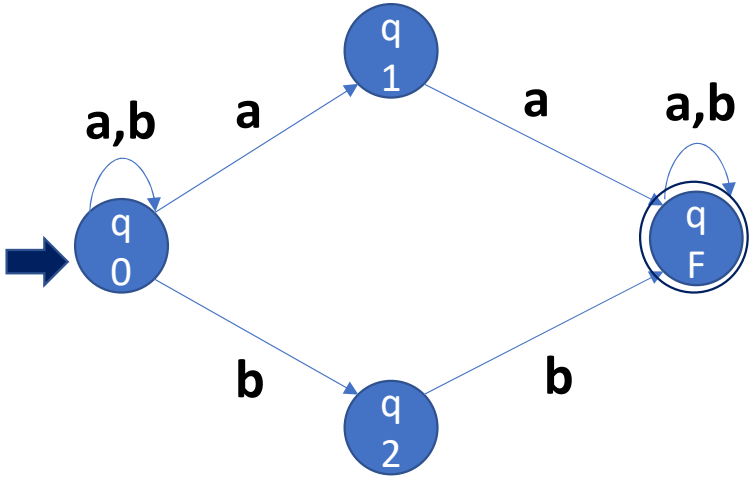
aa

q0

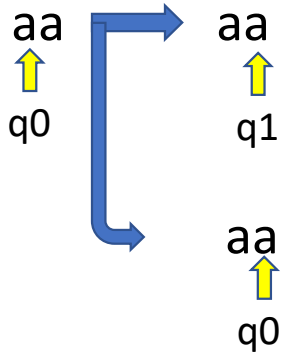


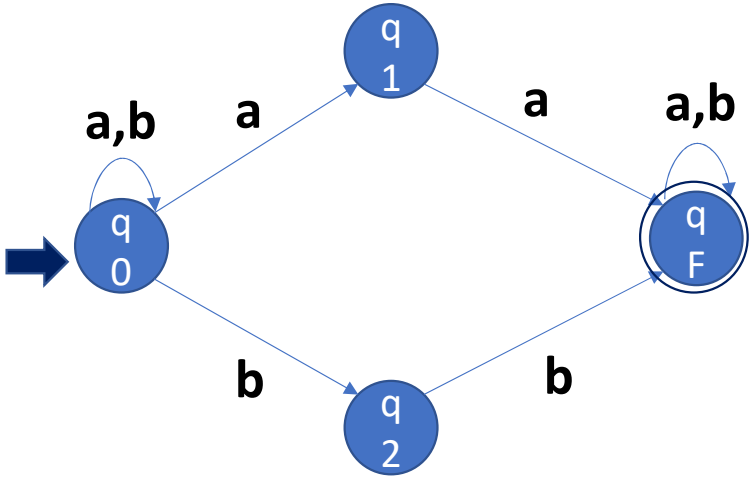
Processamento de palavras:



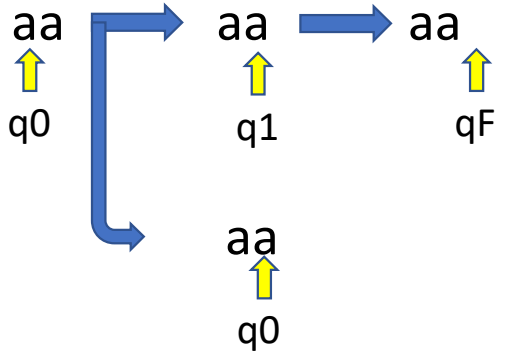


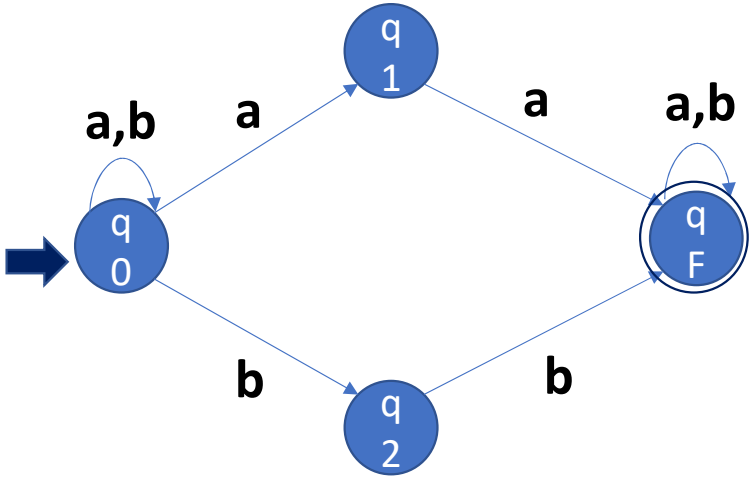
Processamento de palavras:



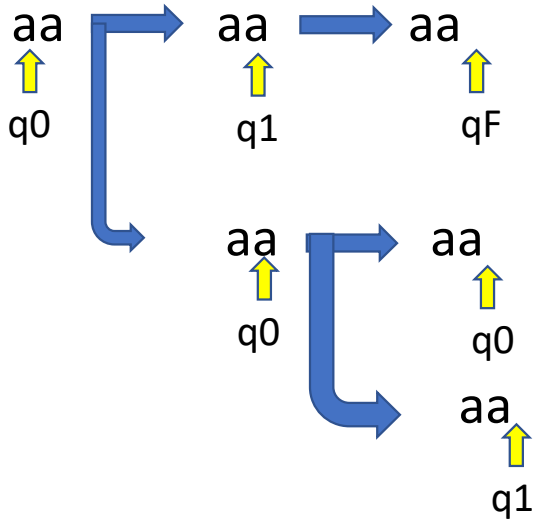


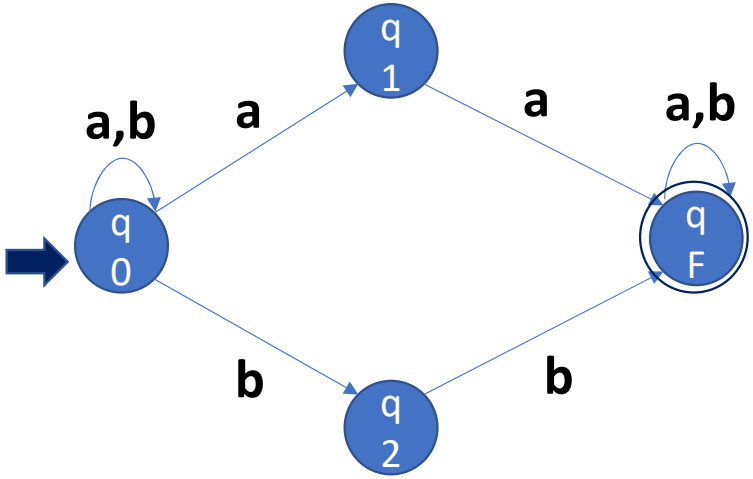
Processamento de palavras:



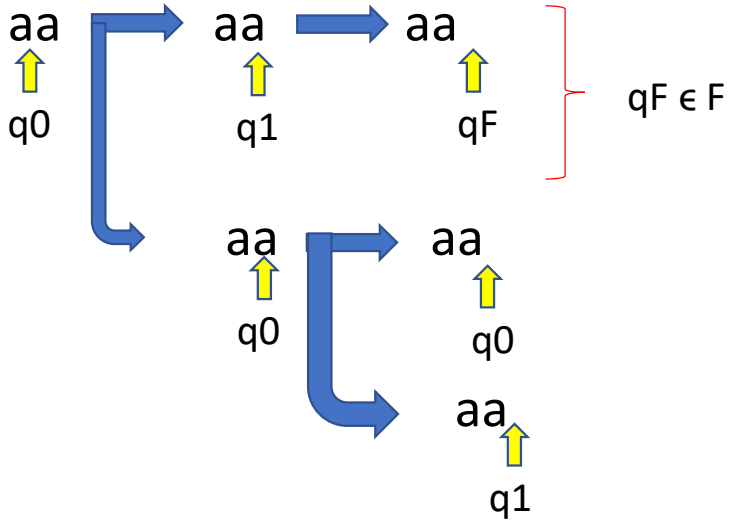


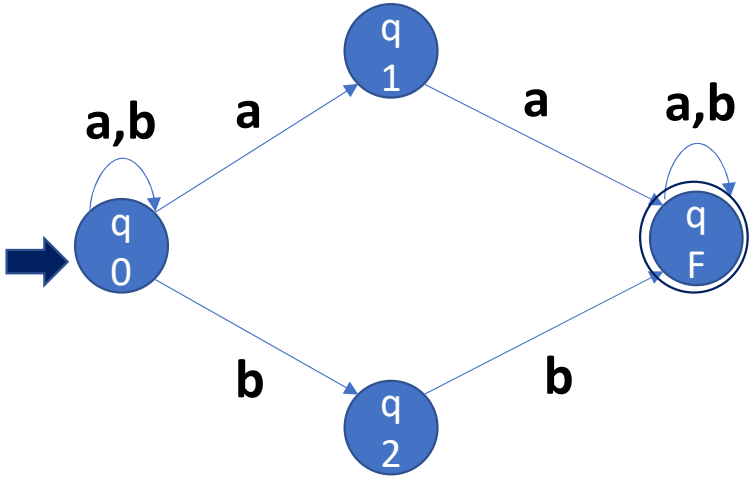
Processamento de palavras:



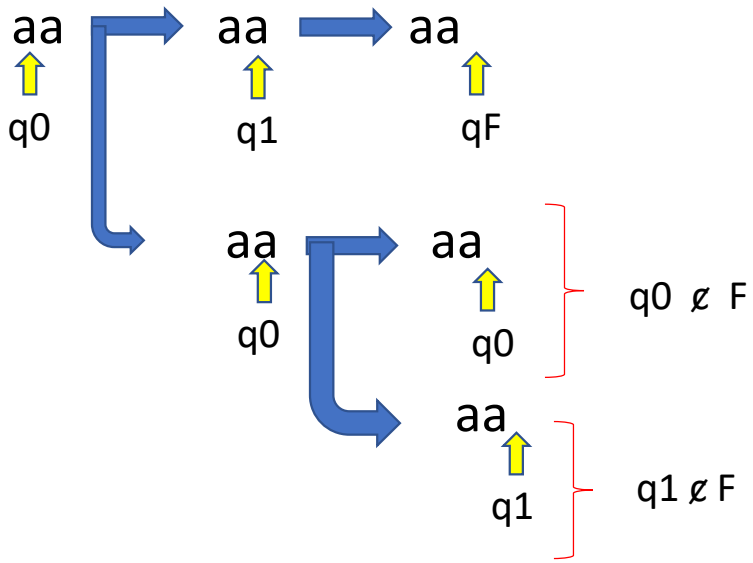


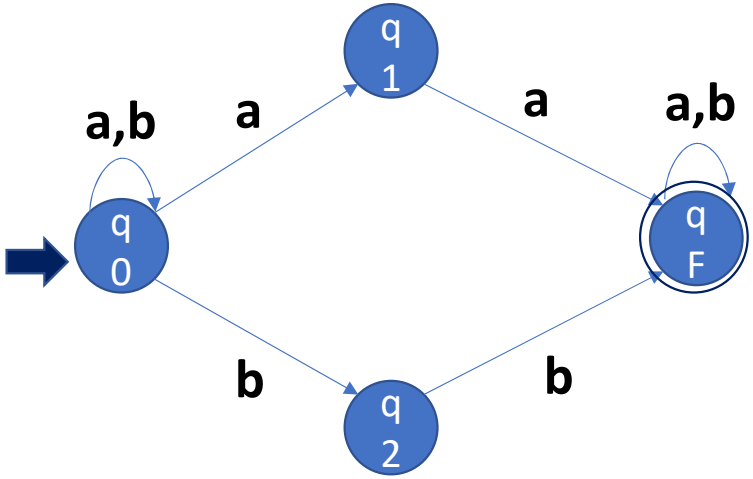
Processamento de palavras:



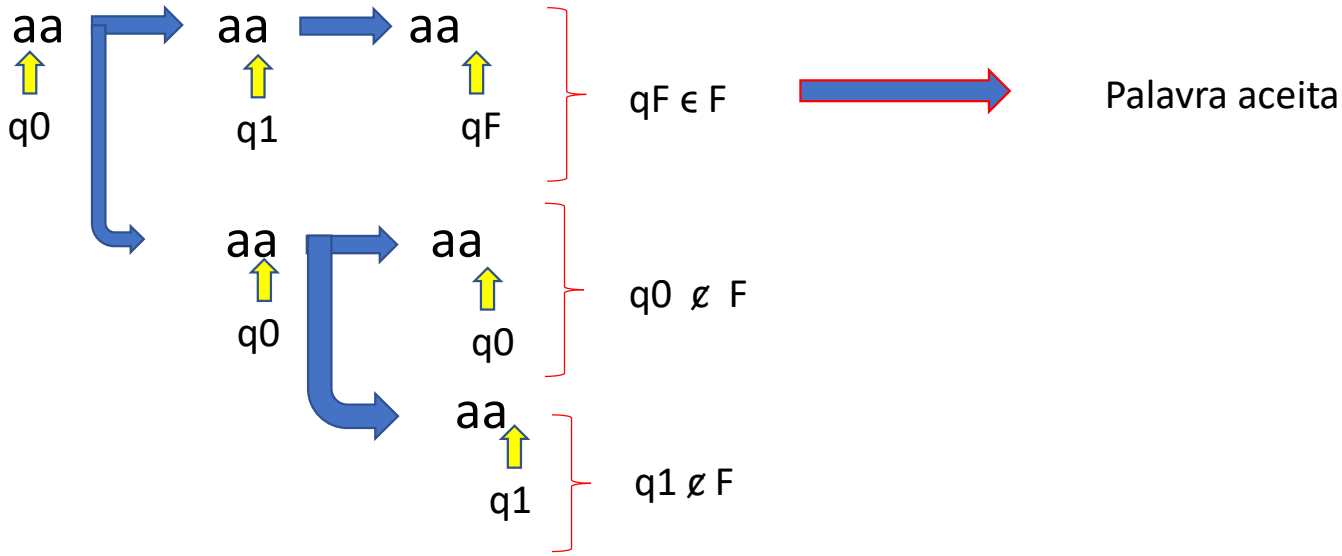


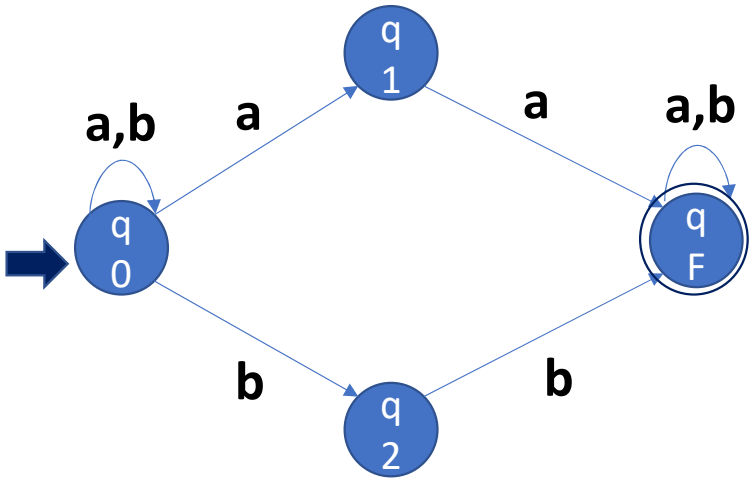
Processamento de palavras:



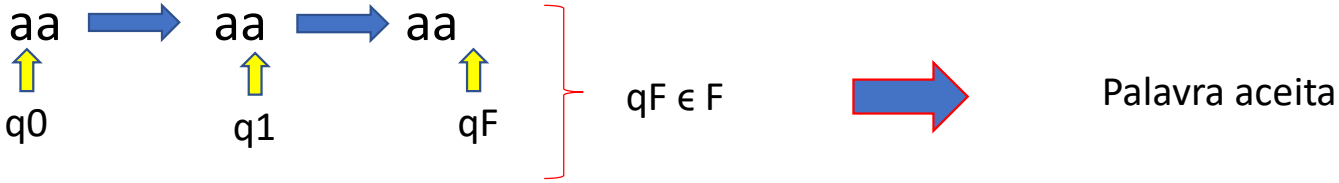


Processamento de palavras:

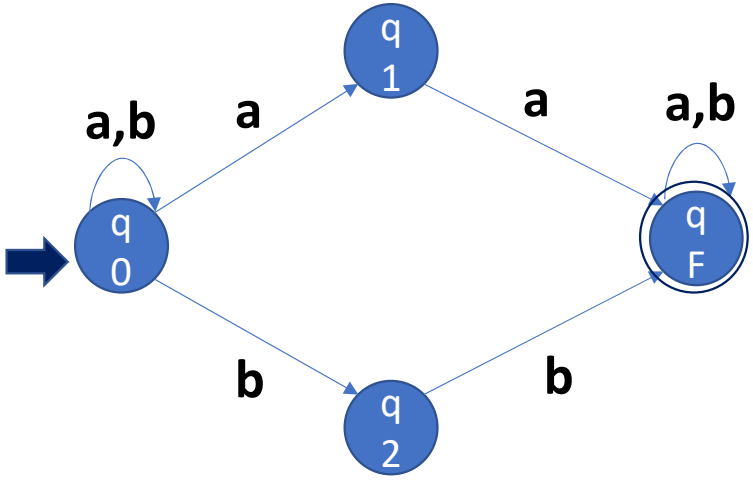




Processamento de palavras:



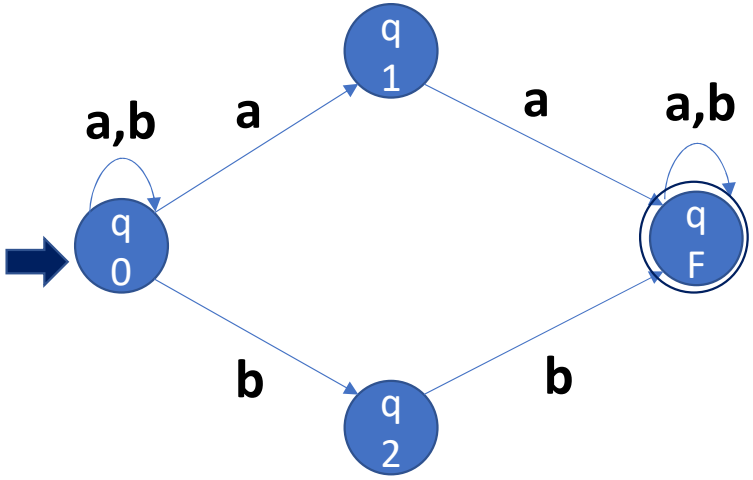
Quando a palavra é aceita, basta apresentar o caminho no qual o processamento leva a um estado final.



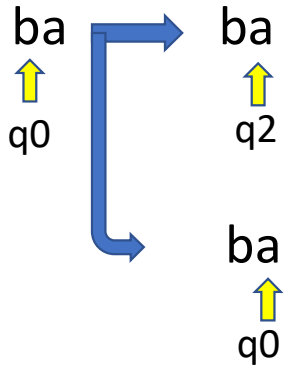
Processamento de palavras:

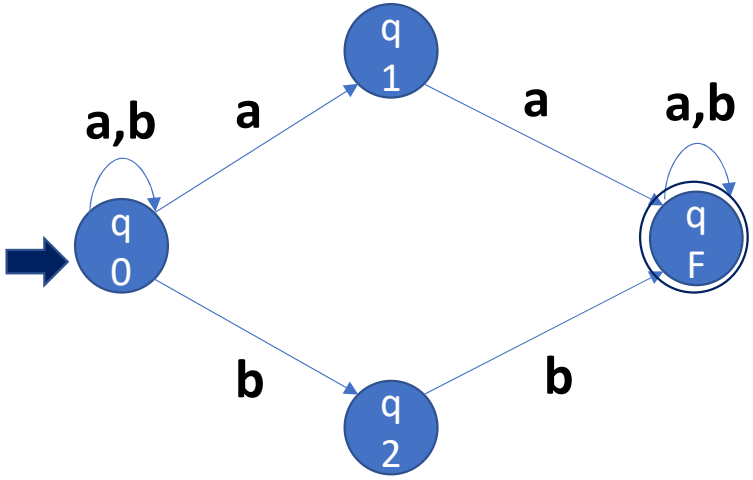
ba

q0

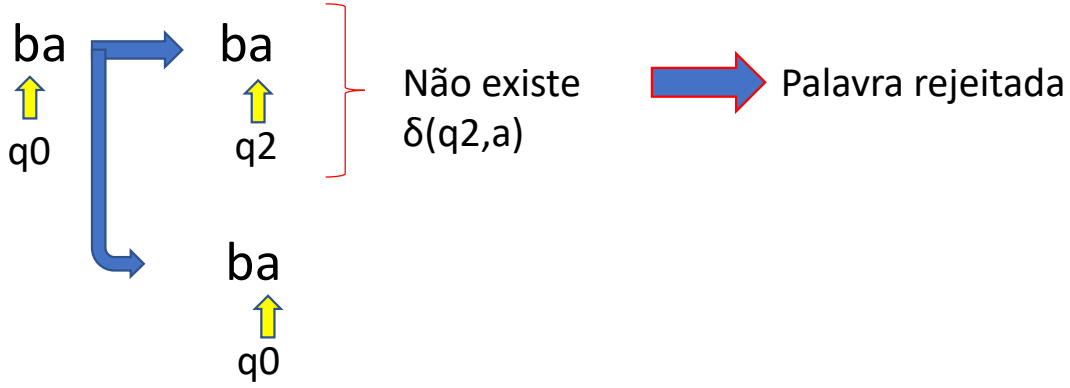


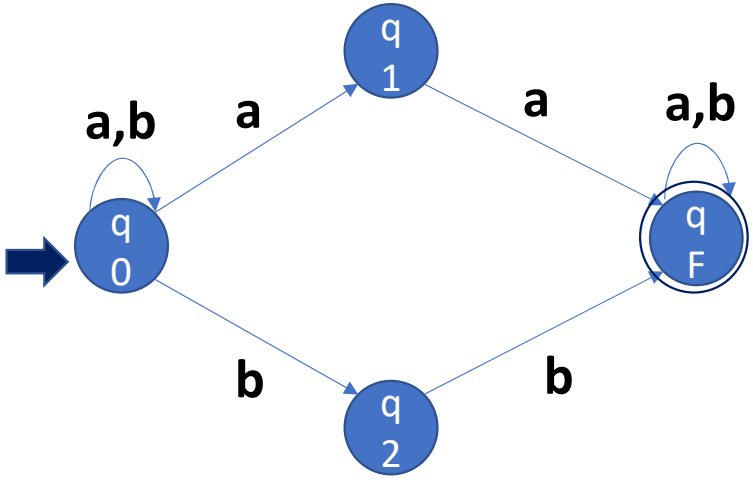
Processamento de palavras:



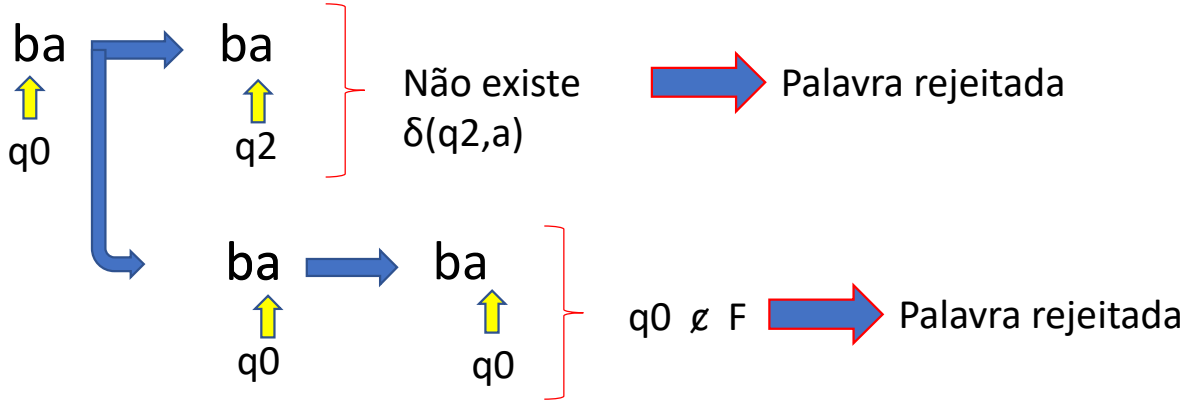


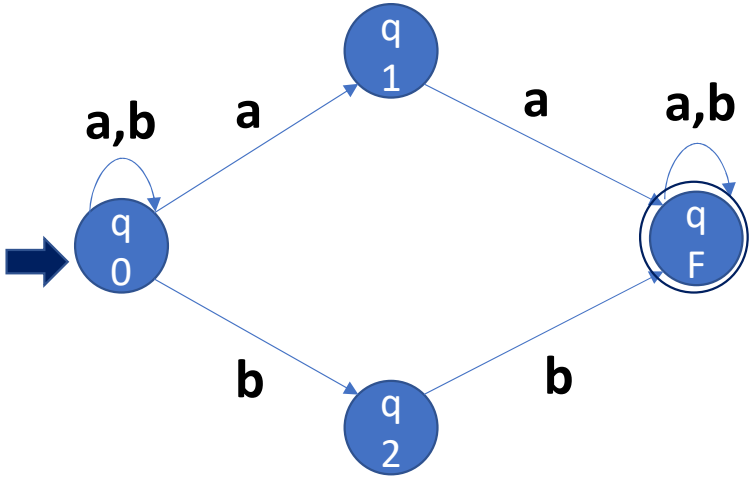
Processamento de palavras:



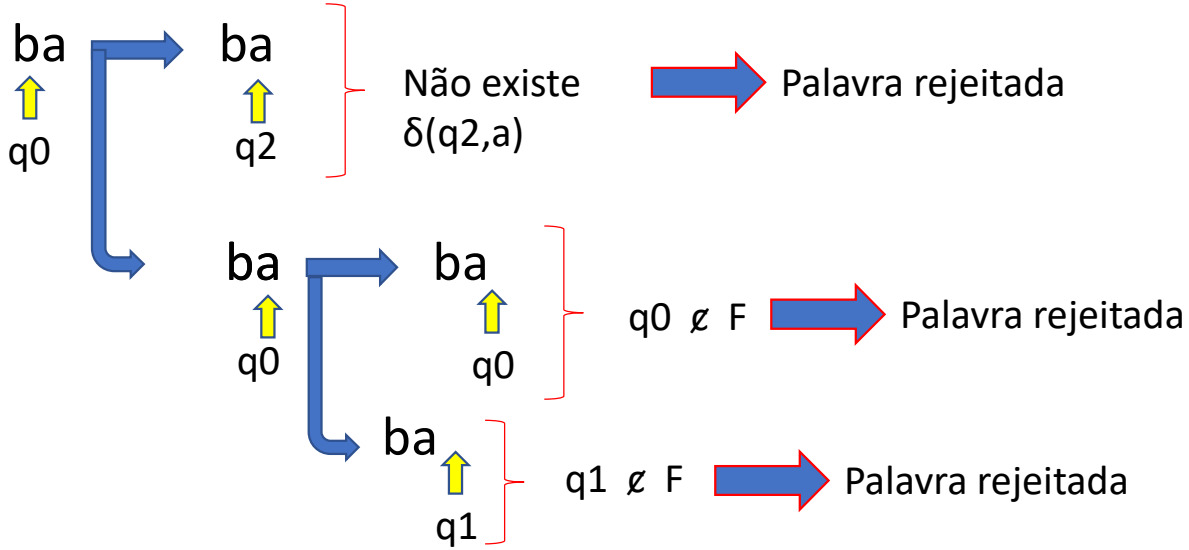


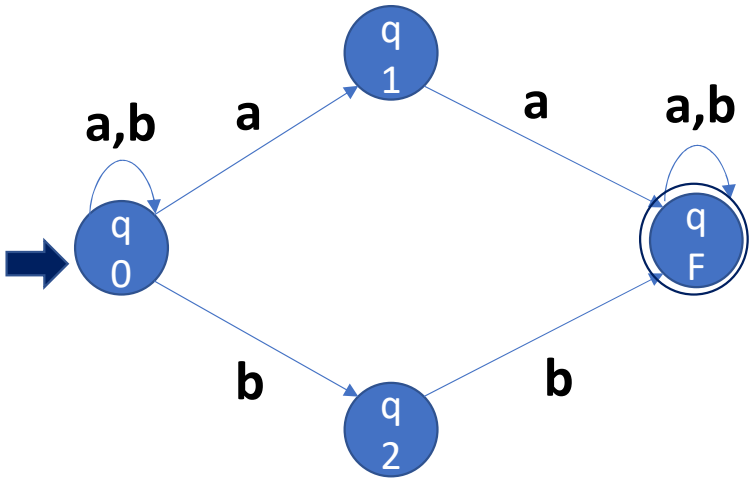
Processamento de palavras:



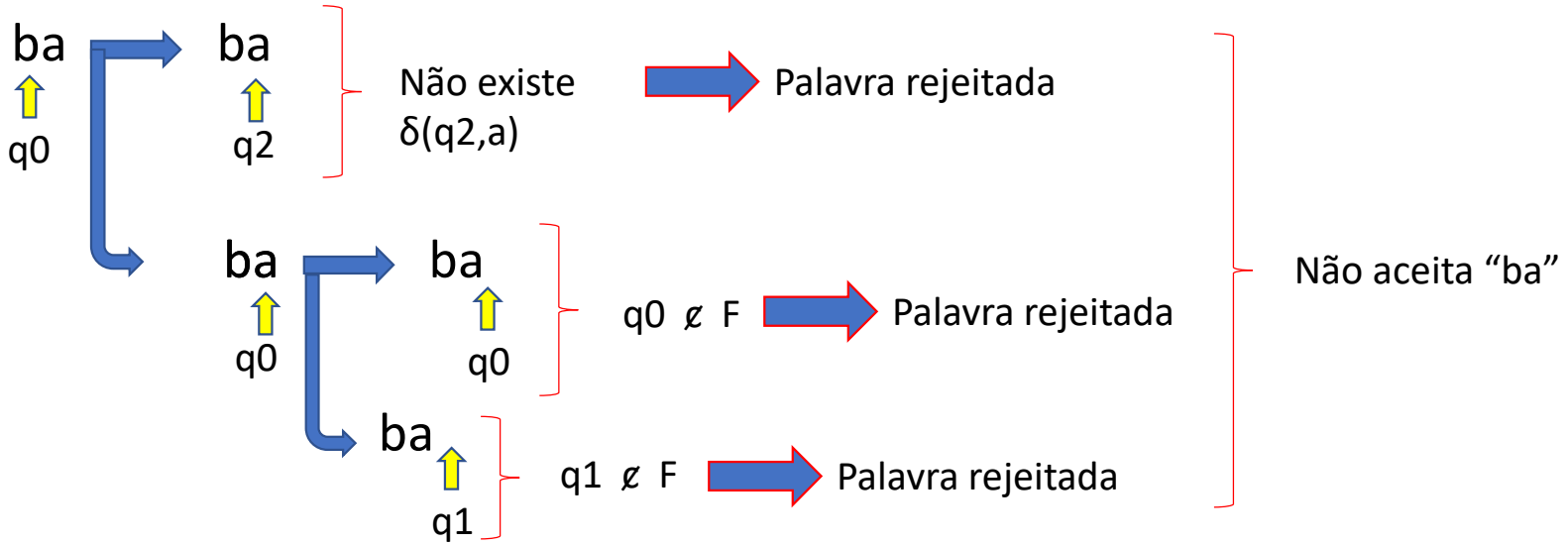


Processamento de palavras:



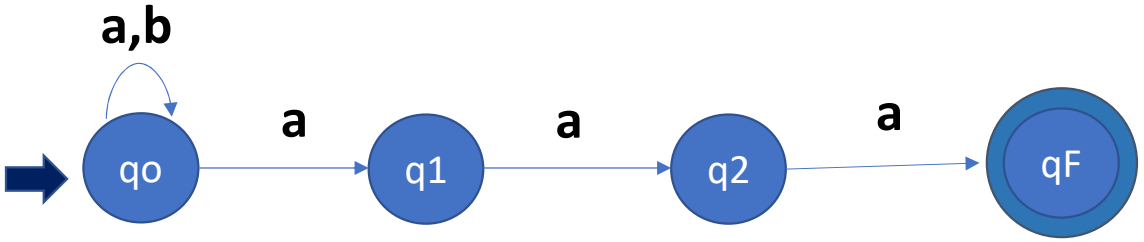


Processamento de palavras:



Exemplo 2:

- $M_6 = (\{a, b\}, \{q_0, q_1, q_2, q_f\}, \delta_6, q_0, \{q_f\})$

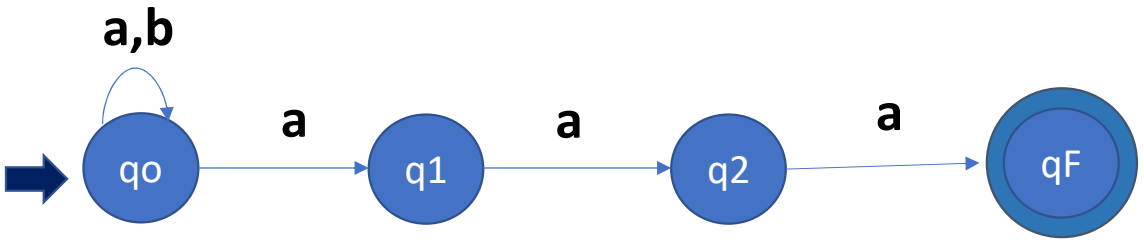


Linguagem reconhecida/aceita por esse AF

L = { w ∈ {a,b}* | “aaa” é sufixo de w}

Exemplo 2:

- $M_6 = (\{a, b\}, \{q_0, q_1, q_2, q_f\}, \delta_6, q_0, \{q_f\})$



L = { w ϵ {a,b}* | “aaa” é sufixo de w}

Exercício 2 (para entregar):

Para o AFND acima, apresentar o processamento das palavras:

a) bababaaa
b) ababaa

♦ Processamento

- união dos resultados da função programa aplicada a cada estado alternativo
- definição formal do comportamento
  - * necessário estender a função programa
  - * argumento: um conjunto finito de estados e uma palavra
- w ∈ ACEITA(M)
  - * pelo menos um caminho alternativo aceita w
- w ∈ REJEITA(M)
  - * todas as alternativas rejeitam w



Definição: Função de Transição Estendida (AFND)

Seja $A = (Q, \sum, \delta, q_0, F)$ um AFND.

A função δ estendida $\underline{\delta}: 2^Q \times \sum^* \rightarrow 2^Q$ estendida para palavra é assim definida:

$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

sendo:

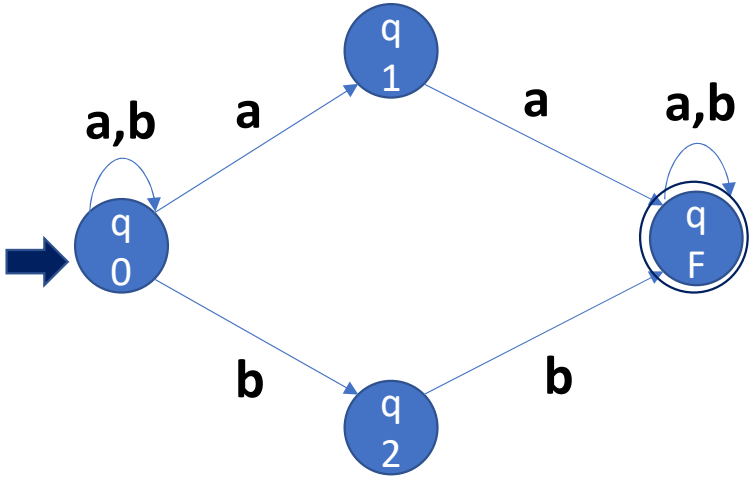
$q \epsilon Q$ (estado)

$S \subset Q$ (subconjunto de estados)

$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)





$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

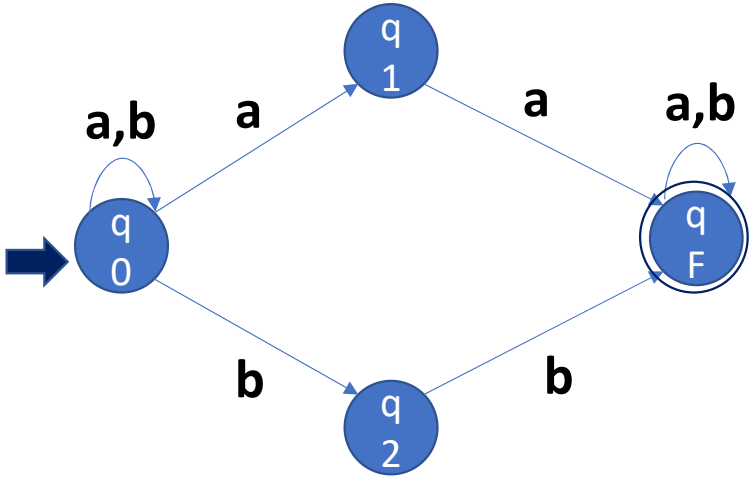
$q \in Q$ (estado)

$S \subset Q$ (subconjunto de estados)

$a \in \sum$ (símbolo)

$w \in \sum^*$ (palavra)

Processamento da palavra “bb”



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)

$S \subset Q$ (subconjunto de estados)

$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)

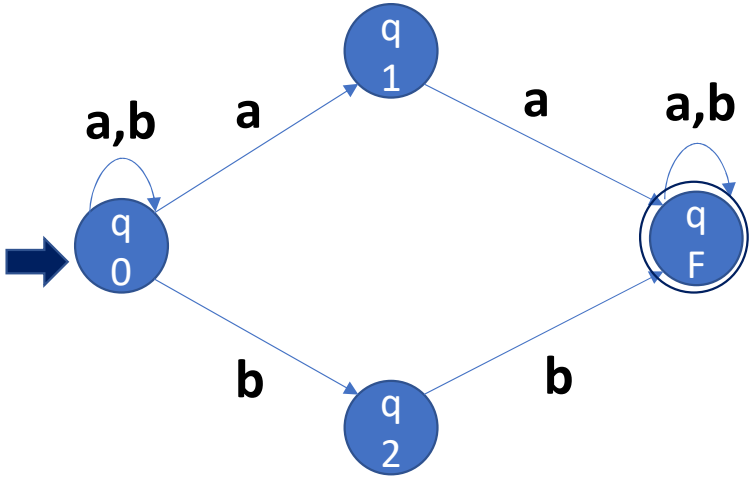
bb



q0

$\underline{\delta}$ (qo,bb) =





$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

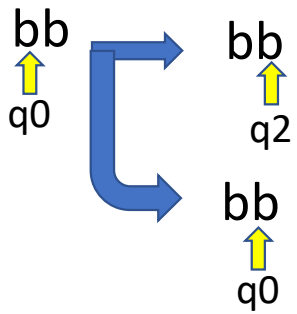
$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)

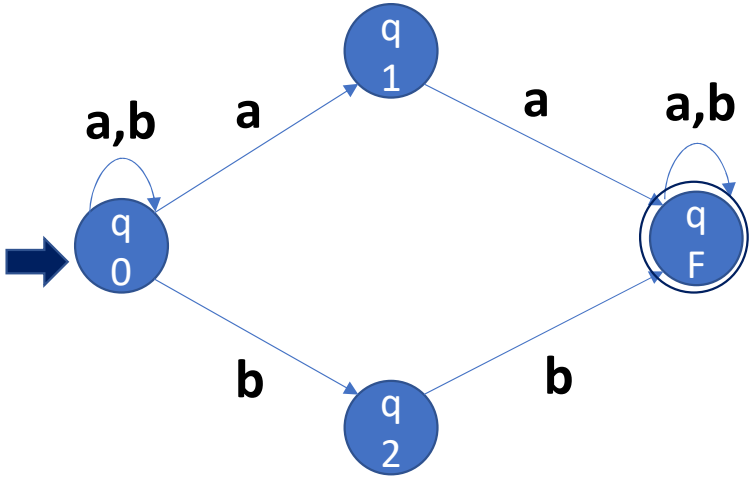
$S \subset Q$ (subconjunto de estados)

$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)

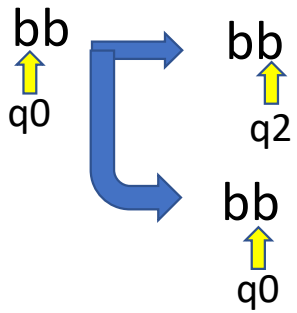


$\underline{\delta}$ (qo,bb) =

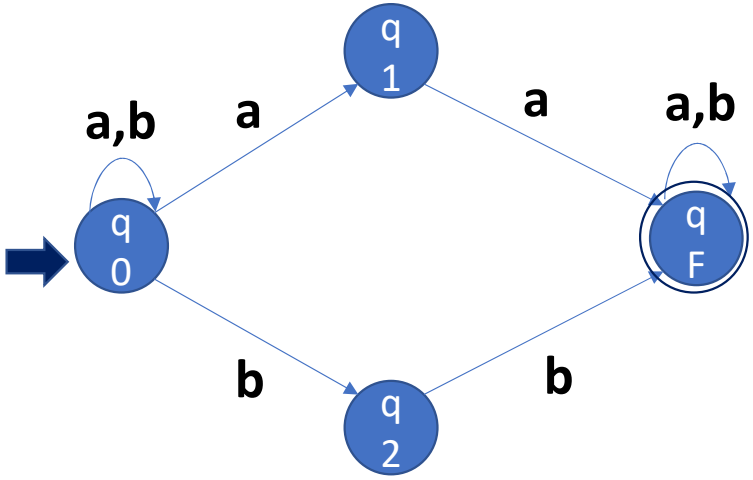


$$\underline{\delta}(q, \varepsilon) = \{q\}$$
$$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$$
$$\underline{\delta}(S, w) = \cup_{q \epsilon S}\, \underline{\delta}(q, w)$$

$q \in Q$ (estado)
$S \subset Q$ (subconjunto de estados)
$a \in \sum$ (símbolo)
$w \in \sum^*$ (palavra)



$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b)



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

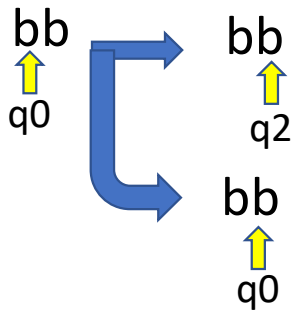
$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)

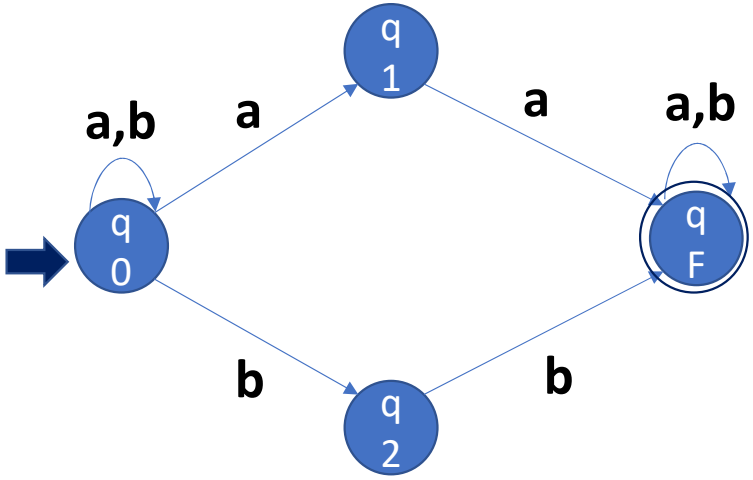
$S \subset Q$ (subconjunto de estados)

$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)

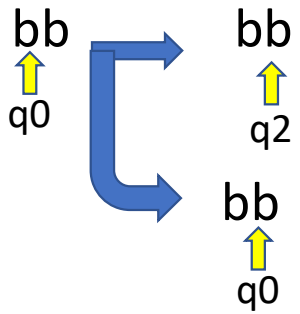


$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b)

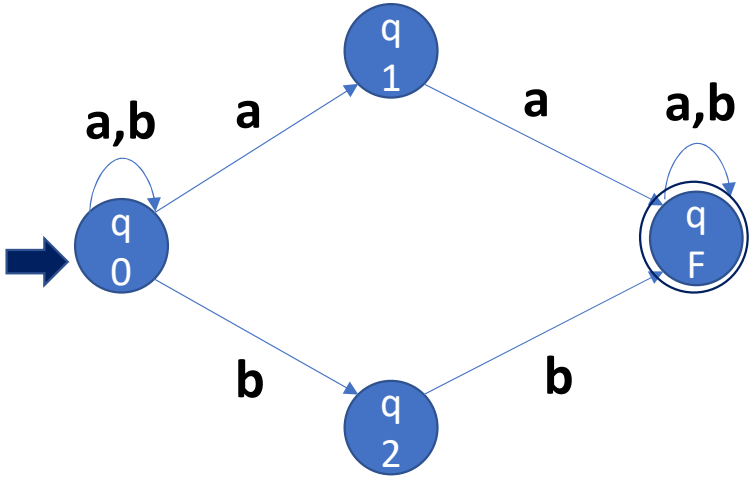


$$\underline{\delta}(q, \varepsilon) = \{q\}$$
$$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$$
$$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$$

$q \in Q$ (estado)
$S \subset Q$ (subconjunto de estados)
$a \in \sum$ (símbolo)
$w \in \sum^*$ (palavra)

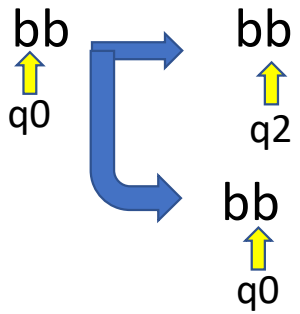


$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b)

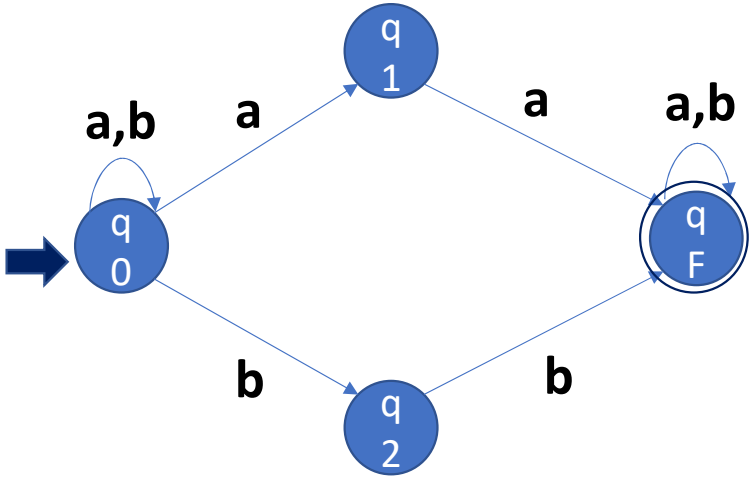


$$\underline{\delta}(q, \varepsilon) = \{q\}$$
$$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$$
$$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$$

$q \in Q$ (estado)
$S \subset Q$ (subconjunto de estados)
$a \in \sum$ (símbolo)
$w \in \sum^*$ (palavra)



$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b)



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

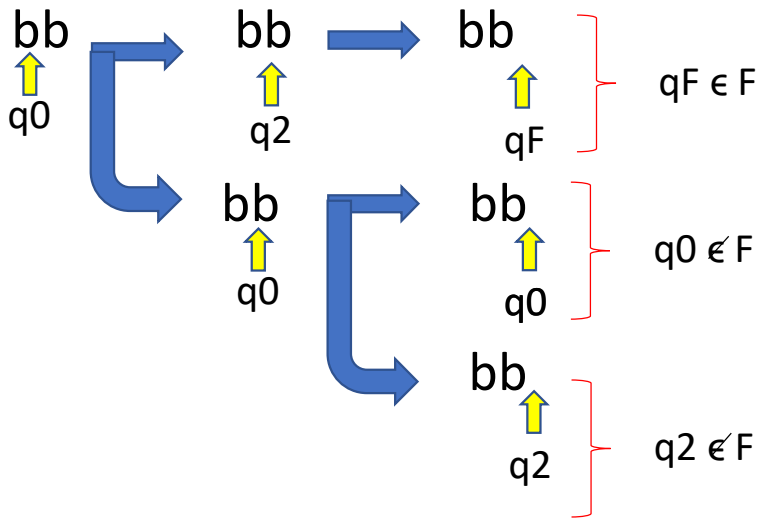
$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)

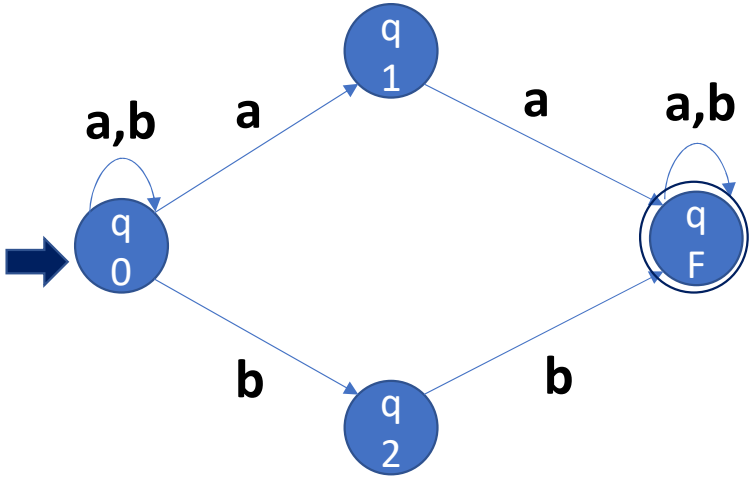
$S \subset Q$ (subconjunto de estados)

$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)



$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) =



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

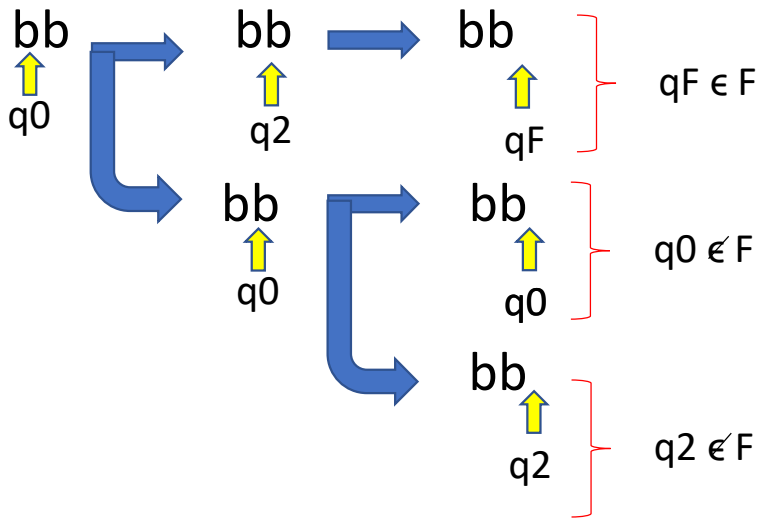
$\underline{\delta}(S, w) = \cup_{q \in S} \underline{\delta}(q, w)$

$q \in Q$ (estado)

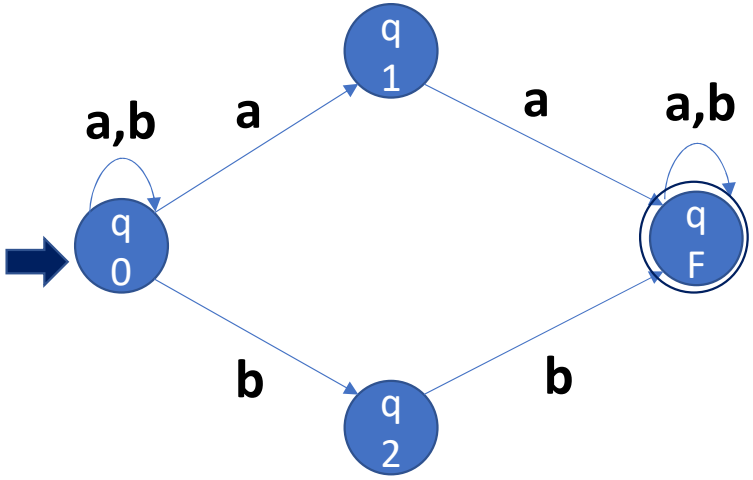
$S \subset Q$ (subconjunto de estados)

$a \in \Sigma$ (símbolo)

$w \in \Sigma^*$ (palavra)



$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b)



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

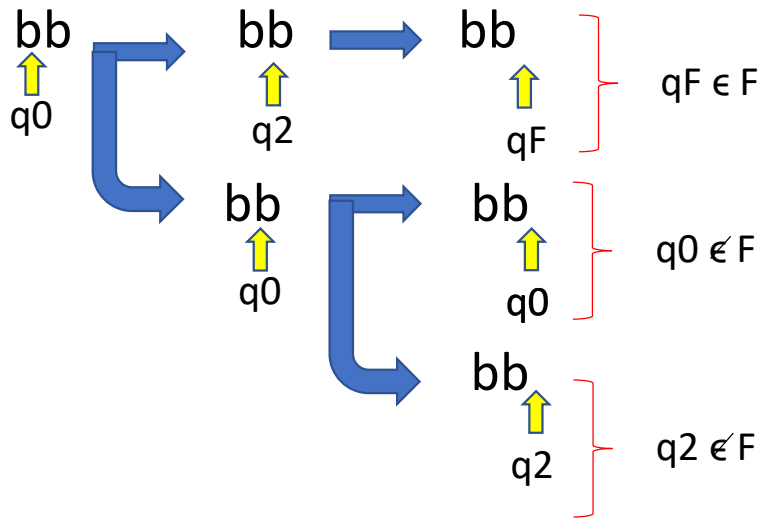
$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)

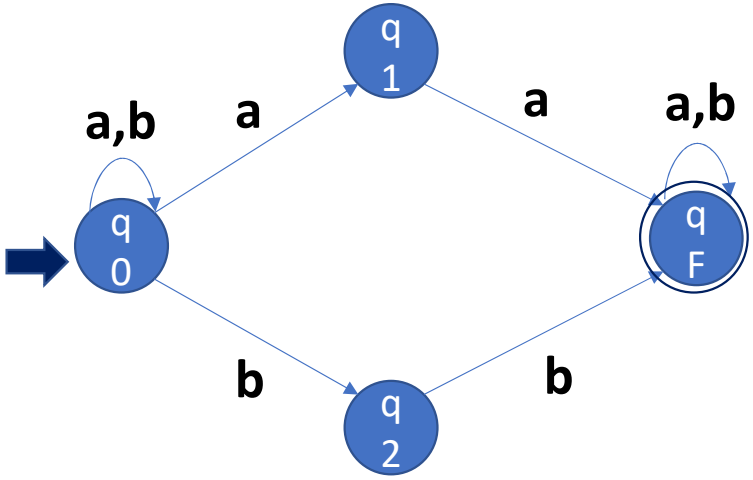
$S \subset Q$ (subconjunto de estados)

$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)



$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ)



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

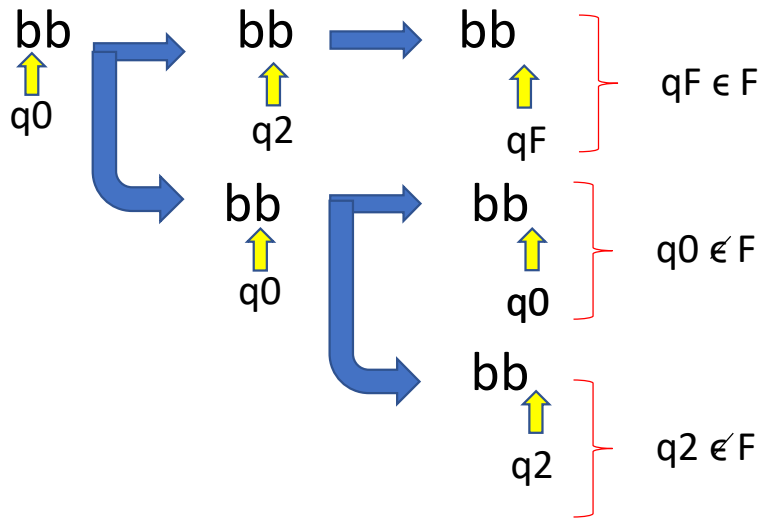
$\underline{\delta}(S, w) = \cup_{q \in S} \underline{\delta}(q, w)$

$q \in Q$ (estado)

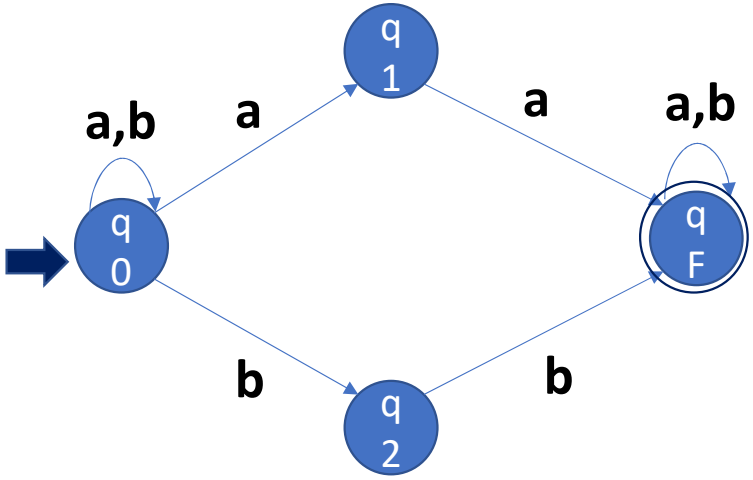
$S \subset Q$ (subconjunto de estados)

$a \in \sum$ (símbolo)

$w \in \sum^*$ (palavra)

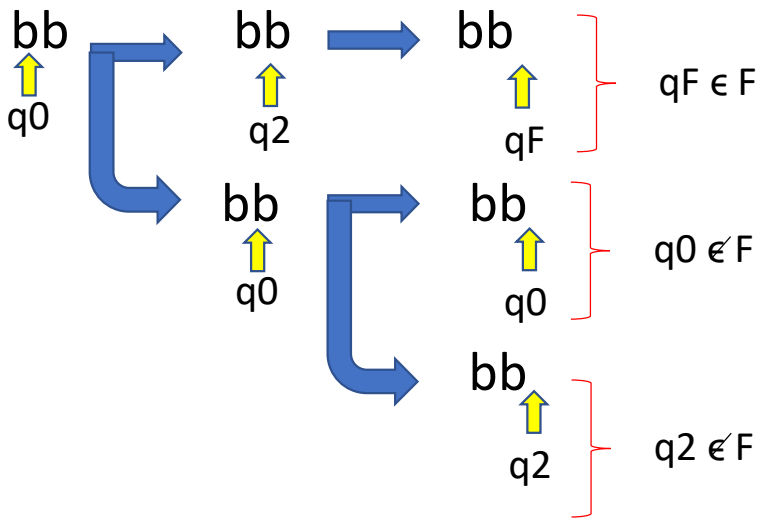


$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ)

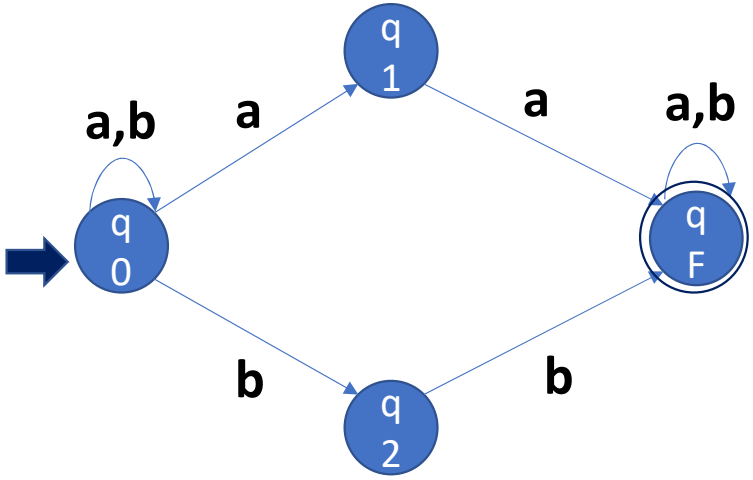


$\underline{\delta}(q, \varepsilon) = \{q\}$
$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$
$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)
$S \subset Q$ (subconjunto de estados)
$a \epsilon \sum$ (símbolo)
$w \epsilon \sum^*$ (palavra)

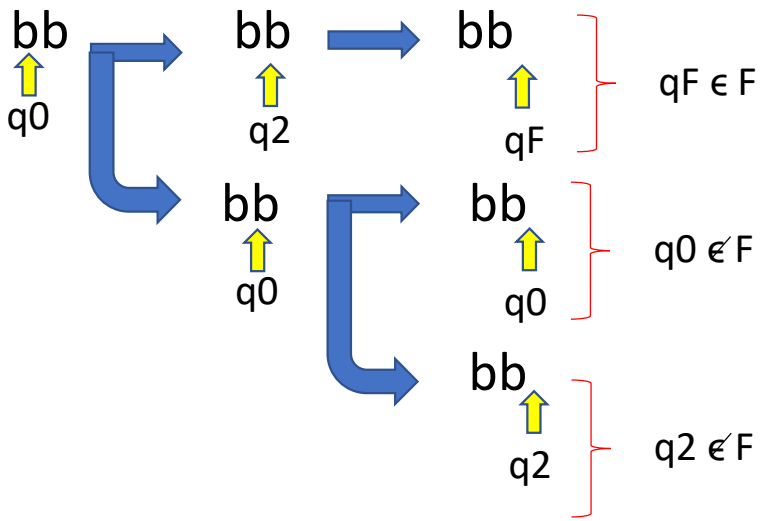


$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ)
= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({qF}, ξ)

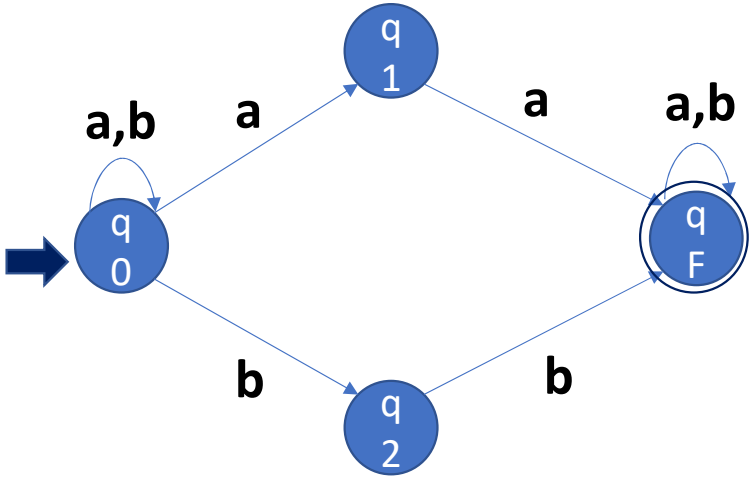


$\underline{\delta}(q, \varepsilon) = \{q\}$
$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$
$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \in Q$ (estado)
$S \subset Q$ (subconjunto de estados)
$a \in \sum$ (símbolo)
$w \in \sum^*$ (palavra)

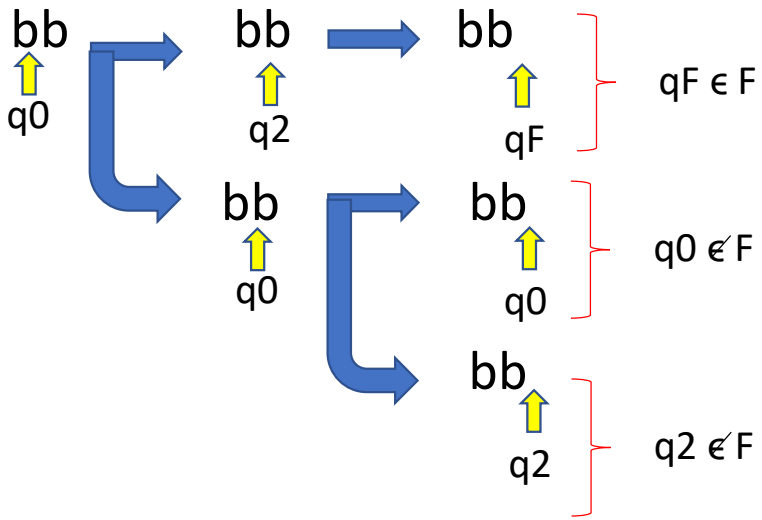


$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ) =
= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({qF}, ξ) = $\underline{\delta}$ (qo, ξ) $\cup$ $\underline{\delta}$ (q2, ξ) $\cup$ $\underline{\delta}$ (qF, ξ)

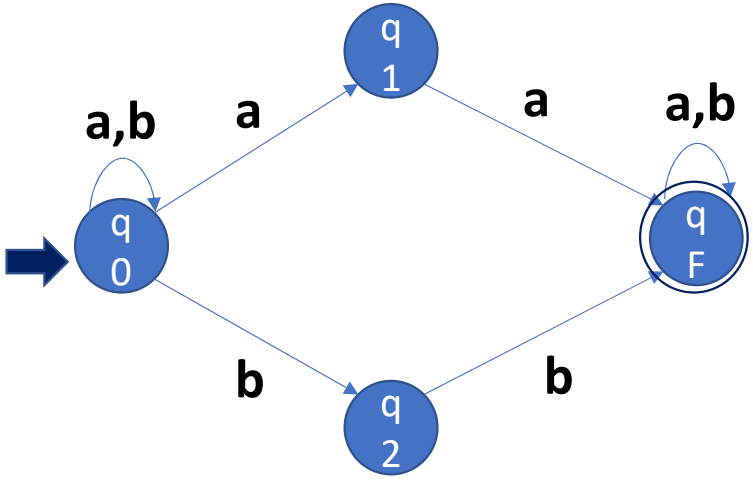


$\underline{\delta}(q, \varepsilon) = \{q\}$
$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$
$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \in Q$ (estado)
$S \subset Q$ (subconjunto de estados)
$a \in \sum$ (símbolo)
$w \in \sum^*$ (palavra)



$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ)
= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({qF}, ξ) = {qo} $\cup$ {q2} $\cup$ {qF}



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

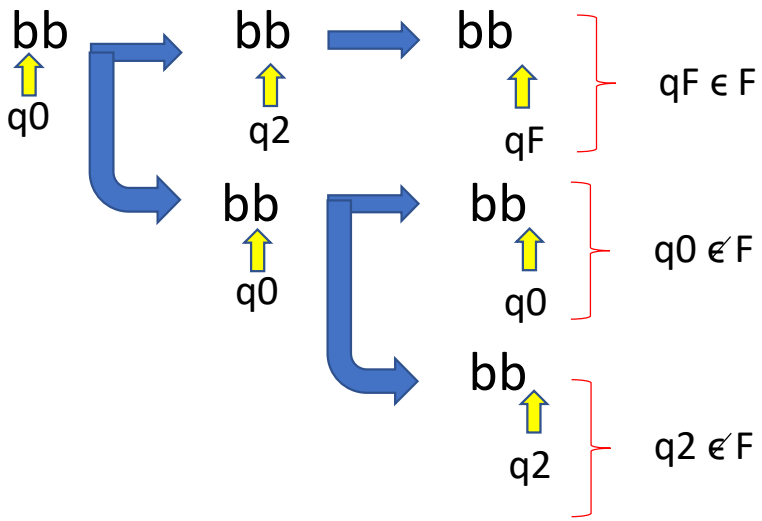
$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)

$S \subset Q$ (subconjunto de estados)

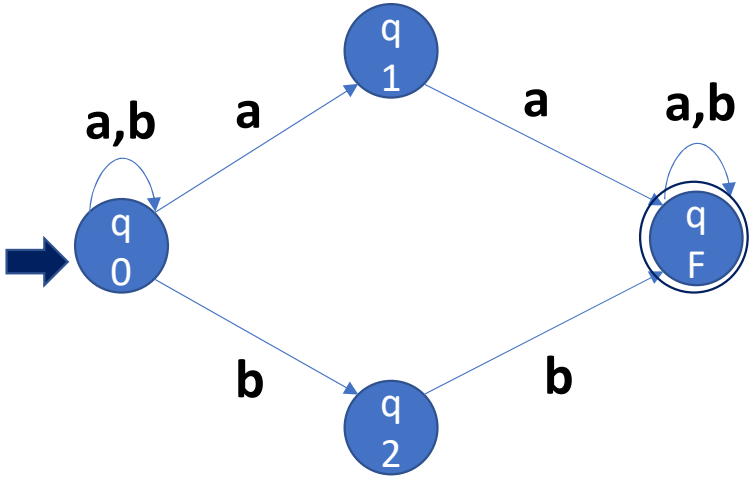
$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)



$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ)

= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({qF}, ξ) = {qo,q2,qF}



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

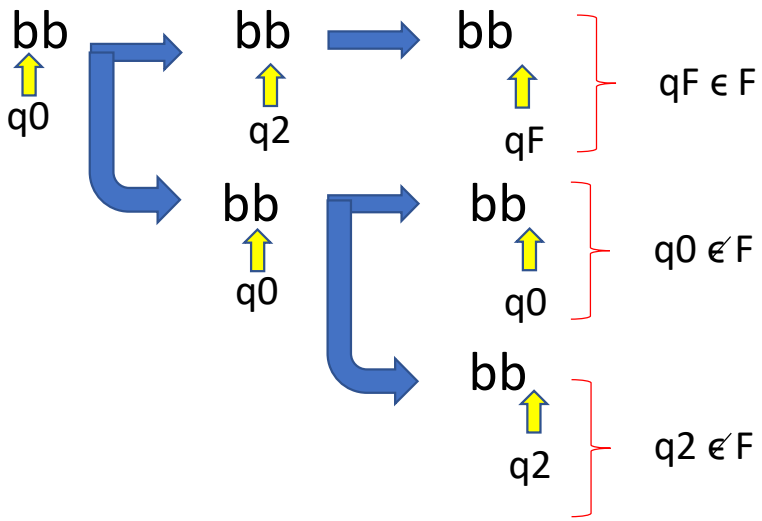
$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)

$S \subset Q$ (subconjunto de estados)

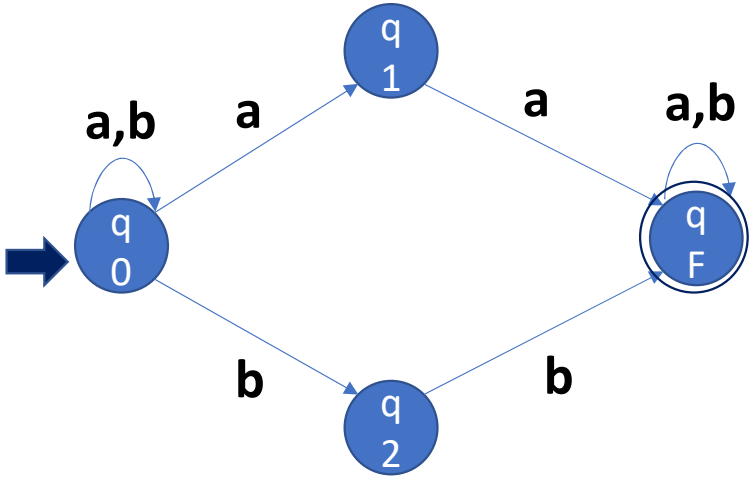
$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)



$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ)

= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({qF}, ξ) = {qo,q2,qF}



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

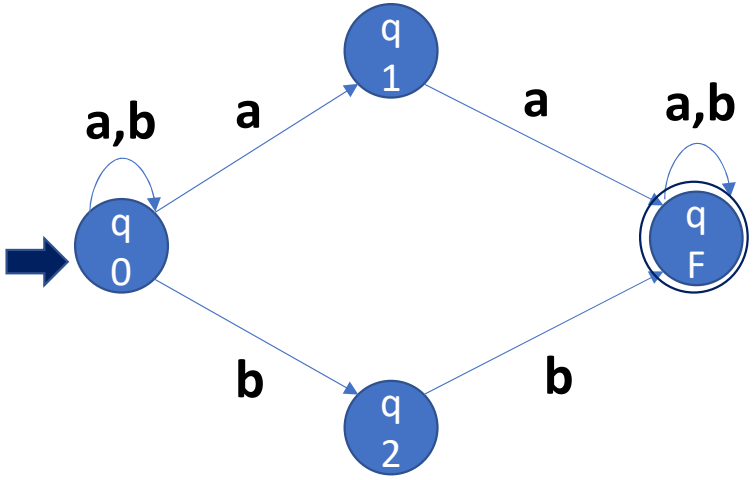
$q \in Q$ (estado)

$S \subset Q$ (subconjunto de estados)

$a \in \sum$ (símbolo)

$w \in \sum^*$ (palavra)

Processamento da palavra “ab”



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)

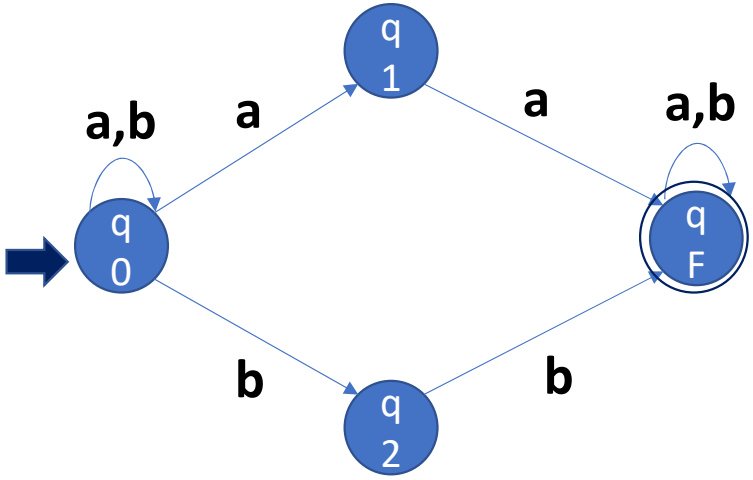
$S \subset Q$ (subconjunto de estados)

$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)

Processamento da palavra “ab”

$\underline{\delta}$ (qo,ab) = $\underline{\delta}$ (δ (qo,a), b)



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \in S} \underline{\delta}(q, w)$

$q \in Q$ (estado)

$S \subset Q$ (subconjunto de estados)

$a \in \sum$ (símbolo)

$w \in \sum^*$ (palavra)

Processamento da palavra “ab”

$\underline{\delta}$ (qo,ab) = $\underline{\delta}$ (δ (qo,a), b) = $\underline{\delta}$ ({q0,q1}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q1, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q1,b), ξ) =

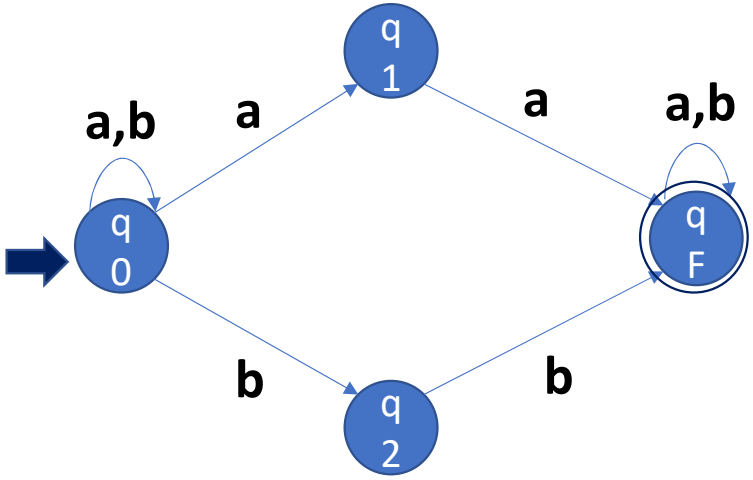
= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({ }, ξ) = {qo,q2}

Definição: Palavra aceita por um AFND

Seja w ϵ $\sum^*$ e o AFND A = (Q, $\sum$, δ, q_0, F).
A palavra w é aceita por A se $\underline{\delta}(q_0, w) \cap F \neq \varnothing$,

ou seja:
se existe algum estado final em $\underline{\delta}(q_0, w)$.





$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \in Q$ (estado)

$S \subset Q$ (subconjunto de estados)

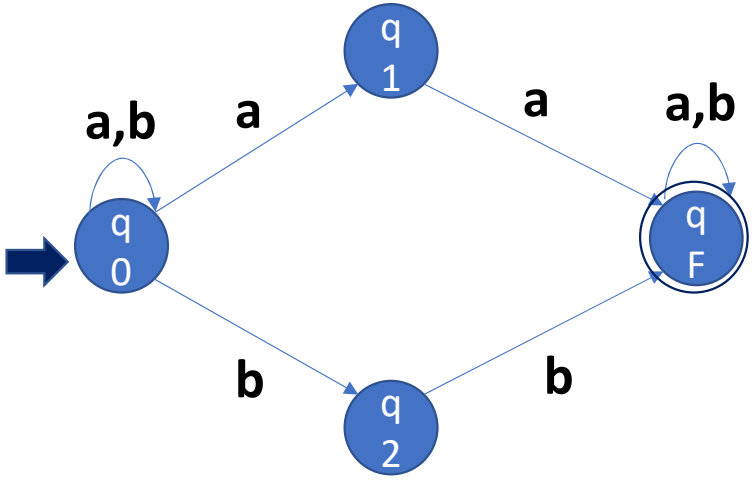
$a \in \sum$ (símbolo)

$w \in \sum^*$ (palavra)

Processamento da palavra “bb”

$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ) =

= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({qF}, ξ) = {qo,q2,qF}



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \in Q$ (estado)

$S \subset Q$ (subconjunto de estados)

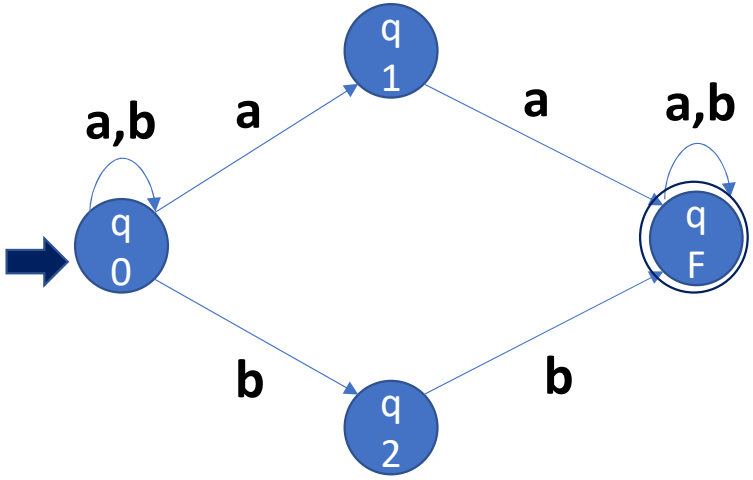
$a \in \sum$ (símbolo)

$w \in \sum^*$ (palavra)

Processamento da palavra “bb”

$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ) =

= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({qF}, ξ) = {qo,q2,qF} $\underline{\delta}(q_0, w) \cap F \neq \varnothing$



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \in Q$ (estado)

$S \subset Q$ (subconjunto de estados)

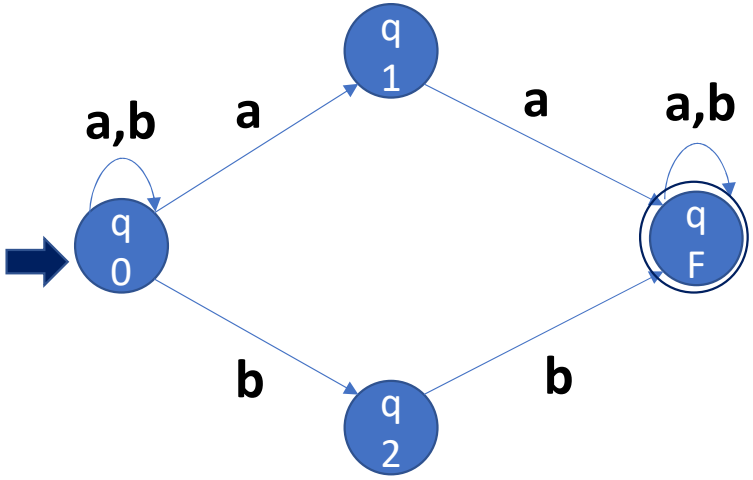
$a \in \sum$ (símbolo)

$w \in \sum^*$ (palavra)

Processamento da palavra “bb”

$\underline{\delta}$ (qo,bb) = $\underline{\delta}$ (δ (qo,b), b) = $\underline{\delta}$ ({q0,q2}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q2, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q2,b), ξ) =

= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({qF}, ξ) = {qo,q2,qF} $\underline{\delta}(q_0, w) \cap F \neq \varnothing$ ➡ palavra “bb” é aceita



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \in S} \underline{\delta}(q, w)$

$q \in Q$ (estado)

$S \subset Q$ (subconjunto de estados)

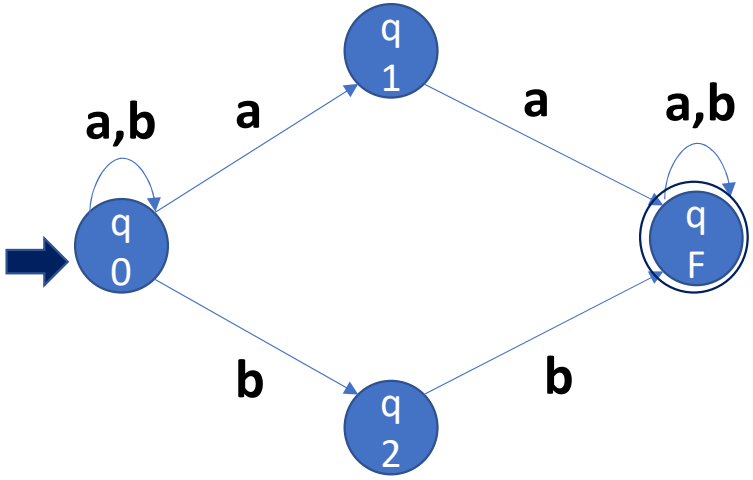
$a \in \sum$ (símbolo)

$w \in \sum^*$ (palavra)

Processamento da palavra “ab”

$\underline{\delta}$ (qo,ab) = $\underline{\delta}$ (δ (qo,a), b) = $\underline{\delta}$ ({q0,q1}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q1, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q1,b), ξ) =

= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({ }, ξ) = {qo,q2}



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \in S} \underline{\delta}(q, w)$

$q \in Q$ (estado)

$S \subset Q$ (subconjunto de estados)

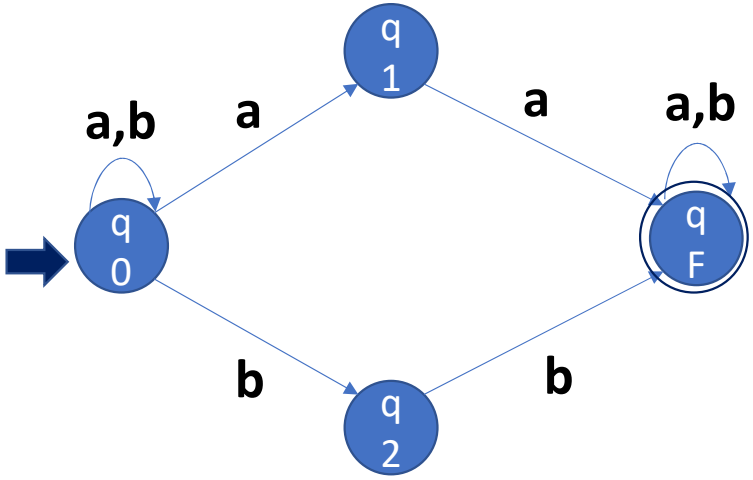
$a \in \Sigma$ (símbolo)

$w \in \Sigma^*$ (palavra)

Processamento da palavra “ab”

$\underline{\delta}$ (qo,ab) = $\underline{\delta}$ (δ (qo,a), b) = $\underline{\delta}$ ({q0,q1}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q1, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q1,b), ξ) =

= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({ }, ξ) = {qo,q2} $\underline{\delta}(q_0, w) \cap F = \varnothing$



$\underline{\delta}(q, \varepsilon) = \{q\}$

$\underline{\delta}(q, aw) = \underline{\delta}(\delta(q, a), w)$

$\underline{\delta}(S, w) = \cup_{q \epsilon S} \underline{\delta}(q, w)$

$q \epsilon Q$ (estado)

$S \subset Q$ (subconjunto de estados)

$a \epsilon \sum$ (símbolo)

$w \epsilon \sum^*$ (palavra)

Processamento da palavra “ab”

$\underline{\delta}$ (qo,ab) = $\underline{\delta}$ (δ (qo,a), b) = $\underline{\delta}$ ({q0,q1}, b) = $\underline{\delta}$ (qo, b) $\cup$ $\underline{\delta}$ (q1, b) = $\underline{\delta}$ (δ (qo,b), ξ) $\cup$ $\underline{\delta}$ (δ (q1,b), ξ) =

= $\underline{\delta}$ ({qo,q2}, ξ) $\cup$ $\underline{\delta}$ ({ }, ξ) = {qo,q2} $\underline{\delta}(q_0, w) \cap F = \varnothing$ ➡ palavra “ab” é rejeitada

Definição: Palavra aceita por um AFND

Seja $w \in \Sigma^*$ e o AFND $A = (Q, \Sigma, \delta, q_0, F)$.
A palavra w é aceita por A se $\underline{\delta}(q_0, w) \cap F \neq \varnothing$,

ou seja:
se existe algum estado final em $\underline{\delta}(q_0, w)$.

Definição: Linguagem aceita por um AFND

A linguagem aceita por A é definida por:

$$L(A) = \{w \in \Sigma^* \mid \underline{\delta}(q_0, w) \cap F \neq \varnothing\}$$



♦ Determinismo × Não-Determinismo

- não-determinismo
  - aparentemente, um significativo acréscimo
  - ao poder computacional de um AF
- na realidade
  - não aumenta o poder computacional
- para cada AFN
  - é possível construir um AFD equivalente
  - (que realiza o mesmo processamento)
  - o contrário também é verdadeiro



Equivalência dentre AFD e AFND

Seja L(A) a linguagem aceita por um AFND A.
Existe um AFD A' que também aceita L(A).

Seja o AFND A = (Q, $\sum$, δ, q0, F).

O AFD A' equivalente a A é dado por A' = (Q', $\sum$, δ', q_0', F'), sendo:

- Q' = 2^Q. Elementos de Q' do tipo $\{q_1, q_2,\ldots,q_k\}$, sendo q_i(i = 1, 2,...,k) ϵ Q, representados por $<q_1\ q_2\ \ldots\ q_k>$
- F' $\rightarrow$ estados de Q' que contêm pelo menos 1 estado de F.
- Q_0' = $\{q_0\}$, representado por $<q_0>$
- $\delta'(<q_1\ q_2\ldots q_j>, a) = <r_1\ r_2\ldots r_i>$ se e somente se $\delta(\{q_1, q_2, \ldots q_j\}, a) = \delta(q_1, a) \cup \delta(q_2, a) \cup \ldots \delta(q_j, a) = \{r_1, r_2,\ldots,r_i\}$,



Prova da equivalência dentre AFD e AFND

- **¿ AFD M' simula AFN M ?**
  - indução no tamanho da palavra
  - deve-se mostrar que

    $\delta'(\langle q_0\rangle, w) = \langle q_1 \ldots q_u\rangle$ sse $\delta(\{q_0\}, w) = \{q_1, \ldots, q_u\}$.
  - *Base da indução.* $|w| = 0$
    - $\delta'(\langle q_0\rangle, \varepsilon) = \langle q_0\rangle$ sse $\delta(\{q_0\}, \varepsilon) = \{q_0\}$
    - V por definição de função programa estendida
  - *Hipótese de indução.* $|w| = n, n \geq 1$
    - suponha que
    - $\delta'(\langle q_0\rangle, w) = \langle q_1 \ldots q_u\rangle$ sse $\delta(\{q_0\}, w) = \{q_1, \ldots, q_u\}$
  - *Passo de Indução.* $|wa| = n + 1$ e $n \geq 1$
    - $\delta'(\langle q_0\rangle, wa) = \langle p_1 \ldots p_v\rangle$ sse $\delta(\{q_0\}, wa) = \{p_1, \ldots, p_v\}$

      o que equivale, por hipótese de indução

      $\delta'(\langle q_1 \ldots q_u\rangle, a) = \langle p_1 \ldots p_v\rangle$ sse $\delta(\{q_1, \ldots, q_u\}, a) = \{p_1, \ldots, p_v\}$

      o que é verdadeiro, por definição de δ'
- **Logo, M' simula M para qq w**

